Pós-Graduação em Ciência da Computação

SISTEMA HÍBRIDO EVOLUCIONÁRIO BASEADO EM DECOMPOSIÇÃO PARA PREVISÃO DE SÉRIES TEMPORAIS

Por

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Tese de Doutorado

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RECIFE

2016

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SISTEMA HÍBRIDO EVOLUCIONÁRIO BASEADO EM DECOMPOSIÇÃO APLICADO A PREVISÃO DE SÉRIES TEMPORAIS

Trabalho apresentado ao Programa de Pós-graduação em Ciência da Computação do Centro de Informática da Universidade Federal de Pernambuco como requisito parcial para obtenção do grau de Doutor em Ciência da Computação.

Orientador: Teresa Bernarda Ludermir

RECIFE

2016

Catalogação na fonte
Bibliotecário Jefferson Luiz Alves Nazareno CRB 4-1758

O48s Oliveira, João Fausto Lorenzato de.
Sistema híbrido evolucionário baseado em decomposição para a previsão de séries temporais / João Fausto Lorenzato de Oliveira. – 2016.
68f.: fig., tab.

Orientadora: Teresa Bernarda Ludermir.
Tese (Doutorado) – Universidade Federal de Pernambuco. CIn. Ciência da Computação, Recife, 2016.
Inclui referências, anexo e apêndice.

1. Inteligência computacional. 2. Inteligência de exames. 3. Previsão de séries temporais. I. Ludermir, Teresa Bernarda (Orientadora). II. Titulo.

006.3 CDD (22. ed.) UFPE-MEI 2016-153

João Fausto Lorenzato de Oliveira

Sistema Híbrido Evolucionário Baseado em Decomposição para Previsão de Séries Temporais

Tese de Doutorado apresentada ao Programa de Pós-Graduação em Ciência da Computação da Universidade Federal de Pernambuco, como requisito parcial para a obtenção do título de Doutora em Ciência da Computação.

Aprovado em: 26/09/2016.

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Dedico esta tese à minha família, amigos e professores, que sempre estiveram presentes e me apoiaram sempre que precisei.

Resumo

A previsão de séries temporais é uma tarefa importante no campo da aprendizado de máquina, possuindo diversas aplicações em mercado de ações, hidrologia, meteorologia, entre outros. A análise da dependência existente nas observações adjacentes da série é necessária para que seja possível prever valores futuros com alguma precisão. Modelos dinâmicos são utilizados para realizar mapeamentos de uma série temporal, se aproximando do mecanismo gerador da série e sendo capazes de realizar previsões. No entanto, o mecanismo gerador de uma série temporal pode produzir padrões lineares e não-lineares que precisam ser devidamente mapeados. Modelos lineares como o auto-regressivo integrado de média móvel (ARIMA) são capazes de mapear padrões lineares, porém não são indicados quando existem padrões não-lineares na série. Já os modelos não-lineares como as redes neurais artificais (RNA) mapeiam padrões não-lineares, mas podem apresentar desempenho reduzido na presença de padrões lineares em relação aos modelos lineares. Fatores como a definição do número de elementos de entrada da RNA, número de amostras de treinamento podem afetar o desempenho. Abordagens híbridas presentes na literatura realizam o mapeamento dos padrões lineares e não-lineares simultaneamente ou aplicando duas ou mais fases nas previsões. Seguindo a suposição de que os modelos são bem ajustados, a diferença entre o valor previsto e a série real demonstra um comportamento de ruído branco, ou seja, considera-se que a diferença entre os valores (resíduo) é composta por choques aleatórios não correlacionados. Na abordagem de duas ou mais fases, o resíduo gerado pelo modelo aplicado na primeira fase é utilizado pelo segundo modelo. O problema do ajuste pode ser decorrente dos parâmetros mal ajustados e também da série temporal devido à possível necessidade de transformações. Tais abordagens geram previsões mais precisas quando comparadas às técnicas tradicionais. Nesta tese, são explorados sistemas evolucionários para a otimização de parâmetros de técnicas lineares e não-lineares visando o mapeamento dos padrões da série temporal. A abordagem proposta utiliza um preprocessamento automático através de um filtro de suavização exponencial para extrair uma série com distribuição normal. A diferença da série temporal e a série filtrada é mapeada por um sistema composto por um método auto-regressivo (AR) e máquina de vetor de suporte para regressão (SVR). Variações do algoritmo de otimização por enxame de partículas (PSO) e algoritmos genéticos são aplicados na otimização dos hiper-parâmetros do sistema. A previsão final é realizada através da soma das previsões de cada série. Para fins de avaliação do método proposto, experimentos foram realizados com bases de problemas reais utilizando métodos da literatura. Os resultados demonstram que o método obteve previsões precisas na maioria dos casos testados. O filtro de suavização exponencial utilizado supõe que a série possua nível constante (sem tendência). Séries que possuem tendências lineares foram devidamente tratadas, no entanto tendências exponenciais ou polinomiais apresentaram desempenho reduzido. O método proposto possui potencial para melhorias, aplicando métodos que

realizem o mapeamento automático de tendências como a suavização exponencial dupla. Nesta tese o método aditivo foi utilizado para combinação de previsões, no entanto em algumas séries o modelo multiplicativo pode ser mais adequado, produzindo previsões mais precisas.

Palavras-chave: Previsão de séries temporais. Sistemas híbridos inteligentes. Sistemas Evolucionários.

Abstract

Time series forecasting is an important task in the field of machine learning and has many applications in stock market, hydrology, weather and so on. The analysis of the dependence between adjacent observations in the series is necessary in order to achieve better forecasts. Dynamic models are used to perform mappings in the time series by approximating to the data generating process and being able to perform predictions.

However, the data generating process of a time series may produce both linear and nonlinear patterns that need to be mapped. Linear models such as the autoregressive integrated moving average (ARIMA) are able to map linear patterns, although not indicated when nonlinear patterns are present in the series. Nonlinear models such as the artificial neural networks (ANNs) perform nonlinear mappings but demonstrate reduced performance in the presence of linear patterns in comparison to linear models.

Hybrid approaches in the literature perform mappings of linear and nonlinear patterns simultaneously or applying two or more phases.Supposing that the models are adjusted to the data, the difference between the predicted value and the data presents a white noise behavior, thus it is considered that the difference of values (residual) is composed by uncorrelated random shocks. In two-phase approaches the residual produced by the linear model in the first phase is used in the nonlinear model. Also the parameters of the models have an important influence on their performance. Such approaches produce more accurate predictions when compared with traditional methods.

In this thesis, we explore evolutionary system in the context of optimization of parameters for both linear and nonlinear methods, taking into consideration the patterns in a time series. In the proposed approach, an exponential smoothing filter is used to decompose a series with normal distribution which is applied to an ARIMA model and the residual series is applied to a system composed by an autoregressive (AR) and a support vector regression methods (SVR). Variations of particle swarm optimization (PSO) algorithm and genetic algorithm (GA) are employed in the optimization of hyper-parameters of the system.

Experiments were conducted using data sets from real world problems comparing with methods in the literature. The results indicate that the method achieved accurate predictions in most cases. The exponential smoothing filter assumes that the given series has no trend patterns. Series with linear trend were detrended, however in series with exponential or polynomial trends the proposed method achieved reduced performance.

The proposed method has potential to improvements by using methods that perform an automatic mapping of trend patterns (double exponential smoothing). In this work, the additive model is adopted, however in some series a multiplicative model could achieve better forecasts.

Keywords: Time series forecasting. Hybrid intelligent systems. Evolutionary Systems.

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Lista de abreviaturas e siglas

ARIMA	Autorregressivo Integrado de Médias Móveis
RNA	Rede Neural Artificial
MA	Médias Móveis
PSO	*Particle Swarm Optimization*
AR	Autorregressivo
GA	*Algoritmo Genético*
SVR	*Support Vector Regression*

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1 Introdução

Uma série temporal pode ser considerada uma sequência de observações medidas com determinada frequência (BOX; JENKINS; REINSEL, 2013), onde os dados podem ser medidos por exemplo a cada hora, semanalmente, anualmente, entre outras frequências. Existem diversas aplicações de séries temporais, porém neste trabalho o foco será na previsão de valores futuros.

A previsão de séries temporais consiste em analisar dados e realizar uma projeção para o futuro. Tomando como base uma série temporal univariada, a previsão será realizada como uma função dos seus valores passados $\hat{y}_{t+1} = f(y_t, y_{t-1}, \ldots, y_{t-p+1}) + \varepsilon_{t+1}$, onde y_t corresponde ao valor da série no instante t, ε_{t+1} representa o erro residual existente em relação ao modelo e p representa o tamanho da dimensão dos dados passados utilizados para realizar a previsão. Zhang *et. al* (ZHANG, 2001a; ZHANG B. EDDY PATUWO, 2001) constatou que a dimensionalidade dos dados de entrada utilizada pelo modelo exerce influência maior nos resultados finais do que quando comparado ao número de neurônios escondidos em redes neurais artificiais (RNA). De acordo com o teorema de Takens (TAKENS, 1981), é possível modelar uma série temporal utilizando uma dimensão suficientemente larga.

Uma técnica amplamente utilizada na previsão de séries temporais é o modelo autorregressivo integrado de médias móveis (ARIMA). O processo de construção do modelo foi proposto por Box e Jenkins (BOX; JENKINS; REINSEL, 2013), onde o componente autorregressivo (AR) é aplicado na série temporal e os resíduos são tratados pelo componente de médias móveis (*moving average*) (MA). Sua principal limitação é assumir linearidade em séries temporais.

As redes neurais artificiais são técnicas não lineares bastante utilizadas em tarefas de previsão de séries temporais devido à sua capacidade de gerar modelos menos sensíveis a dados com ruído (ZHANG, 2001b). No entanto, seu desempenho em séries temporais com características lineares demonstra resultados variados. Markham e Rakes (MARKHAM; RAKES, 1998) compararam RNAs e regressão linear em problemas lineares de regressão com variação no ruído e no número de exemplos da base. RNAs obtiveram resultados superiores em bases com maior variância, enquanto que em bases com menor variância, o método de regressão linear obteve melhores resultados. No campo de estudo das séries temporais, Zhang (ZHANG, 2001a) aplicou RNAs em dados lineares e concluiu que a presença de ruído em séries temporais pode favorecer a utilização de RNAs sobre os modelos lineares.

O processo que gera uma determinada série temporal pode conter características

lineares ou não lineares. Em aplicações do mundo real é comum a ocorrência de ambos os padrões e é difícil determinar o grau de linearidade do processo (ZHANG, 2003). A aplicação de modelos ARIMA em séries não lineares pode não ser adequada. Na literatura, as RNAs apresentam desempenho variado quando aplicadas a problemas lineares (ZHANG, 2003) logo, não é recomendada a utilização de redes neurais em qualquer tipo de dados sem a devida análise. Os experimentos realizados em (ZHANG, 2003) contemplavam séries temporais com número de amostras variando entre 50 a 200, número de neurônios variando entre 1 a 10, e dimensão dos dados de entrada entre 1 a 5.

As máquinas de vetor de suporte para regressão (SVR) também são técnicas não lineares e aplicam uma otimização convexa dos seus parâmetros, ou seja o problema é formulado de forma a só haver um mínimo local para os valores de hiper-parâmetros fornecidos. No entanto, o SVR também possui sensibilidade aos valores de hiper-parâmetros escolhidos.

Na literatura, a combinação de métodos foi proposta como uma forma de mapear differentes padrões em séries temporais obtendo resultados superiores em comparação com as técnicas individuais. Zhang (ZHANG, 2003), Pai e Lin (PAI; LIN, 2005), Xuemei *et al.* (XUEMEI et al., 2010), Zhu e Wei (ZHU; WEI, 2013) e Babu e Reddy (BABU; REDDY, 2014) assumem que uma série temporal pode ser considerada como a soma de padrões lineares e não lineares como é mostrado na Equação (1.1), onde Y_t é um vetor contendo todos os valores da série temporal, L_t e N_t correspondem aos componentes lineares e não lineares respectivamente.

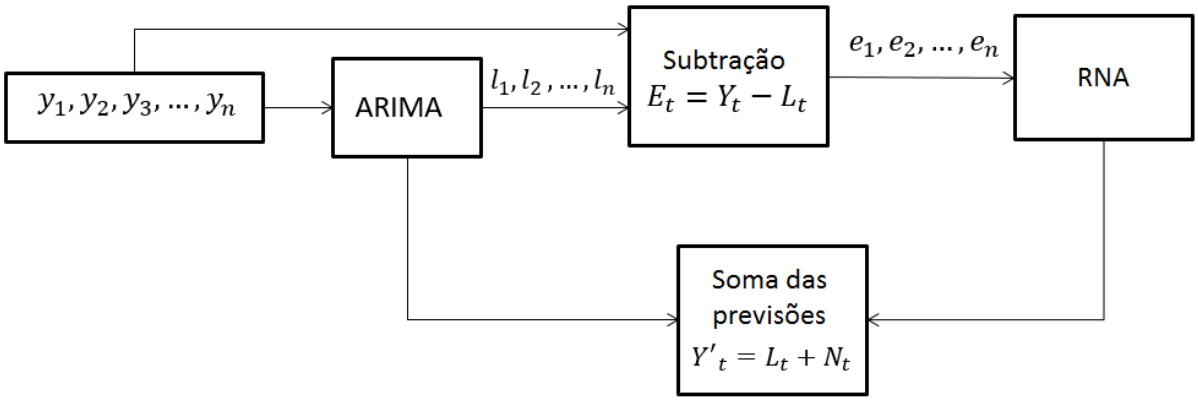
$$Y_t = L_t + N_t. \tag{1.1}$$

Na abordagem de Zhang (ZHANG, 2003) mostrada na Figura 1, uma RNA é utilizada na modelagem do componente não linear, enquanto que um modelo ARIMA é aplicado no componente linear. Essa abordagem também pode ser classificada como de duas fases, onde na primeira o modelo ARIMA é aplicado diretamente nos dados e os resíduos produzidos são modelados pela técnica não linear.

O modelo ARIMA por definição é ajustado de tal forma que os resíduos (diferença entre a série e as previsões) produzidos sejam choques aleatórios estatisticamente independentes com distribuição normal e variância constante (BOX; JENKINS; REINSEL, 2013). O método de modelagem Box-Jenkins (BOX; JENKINS; REINSEL, 2013) é utilizado em séries estacionárias, portanto sua estacionaridade deve ser verificada através de funções de autocorrelação.

A identificação do modelo ARIMA por meio de ferramentas gráficas pode gerar um modelo mal ajustado, contudo esse mesmo modelo pode ser utilizado no método híbrido. De acordo com Granger (BATES; GRANGER, 1969) cada componente do sistema híbrido

Figura 1 – Método de Zhang



Fonte: (ZHANG, 2003).

pode não ser o melhor possível. Abordagens de uma fase mapeiam os padrões lineares e não lineares simultaneamente. Exemplos dessa abordagem são os métodos propostos por Yolcu *et. al* (YOLCU; EGRIOGLU; ALADAG, 2013) e Mohammadzaheri *et. al* (MOHAMMADZAHERI et al., 2009).

Sistemas híbridos compostos por mais de um modelo podem ser capazes de reduzir a variância do sistema (BATES; GRANGER, 1969), no entanto a especificação empírica de cada modelo pode ter desempenho reduzido em relação a acurácia das previsões. Tendo em vista o aumento na precisão de cada técnica em relação às previsões, algumas metodologias híbridas da literatura aplicaram algoritmos de otimização para encontrar um conjunto de valores de hiper-parâmetros das técnicas de forma automática.

Um algoritmo de otimização é o *particle swarm optimization* (PSO) (KENNEDY; EBERHART, 1995), que se inspira em interações sociais de pássaros para resolução de problemas. O PSO tem sido utilizado em diversas aplicações de séries temporais. Yolcu *et. al* (YOLCU; EGRIOGLU; ALADAG, 2013) utilizou o PSO para encontrar os pesos do sistema de redes neurais. Alwee *et al.* (ALWEE; SHAMSUDDIN; SALLEHUDDIN, 2013) utilizaram o PSO para encontrar os hiper-parâmetros de um sistema híbrido composto por um modelos ARIMA e *support vector regression* (SVR) (DRUCKER et al., 1997), Oliveira e Ludermir (OLIVEIRA; LUDERMIR, 2014) utilizaram PSO na otimização de um sistema ARIMA-SVR fazendo seleção de *lags* nas séries temporais. Oliveira e Ludermir (OLIVEIRA; LUDERMIR, 2016) aplicaram um filtro de suavização exponencial para produzir uma série Gaussiana e outra não Gaussiana, onde cada componente foi modelado por um modelo ARIMA e um modelo composto por um modelo AR-SVR. PSO foi utilizado para a otimização dos hiper-parâmetros do sistema.

Este trabalho trata da utilização de algoritmos de otimização em sistemas de previsão de séries temporais, considerando as metodologias de decomposição. Além da análise do desempenho das técnicas propostas, a influência de cada componente do sistema

híbrido é investigada.

O restante deste capítulo está organizado da seguinte forma: A Seção 1.1 apresenta a motivação para os estudos em decomposição de séries temporais em termos de padrões lineares e não lineares. A Seção 1.2 traz os principais objetivos deste trabalho. A Seção 1.3 apresenta a produção bibliográfica gerada durante o desenvolvimento desta tese. A organização deste documento é descrita na Seção 1.4.

1.1 MOTIVAÇÃO

A decomposição de séries temporais em termos de padrões lineares e não lineares se tornou uma estratégia popular para a previsão de séries temporais. Diversos trabalhos adotam essa abordagem (ZHANG, 2003; PAI; LIN, 2005; KHASHEI; BIJARI, 2011) devido aos seguintes motivos:

- Dificuldade de determinar se a série temporal é gerada por um processo linear ou não linear (ZHANG, 2003).
- Séries do mundo real geralmente contém ambos os padrões lineares e não lineares (KHASHEI; BIJARI, 2011).
- Nenhum método consegue o melhor desempenho em todas as situações (KHASHEI; BIJARI, 2011).

O primeiro item reflete o problema na seleção do modelo para previsão, onde a combinação de modelos lineares e não lineares seria mais adequada em termos práticos, pois pode ser difícil determinar todas as características dos dados oriundos de problemas reais (ZHANG, 2003). No segundo item, a abordagem híbrida é projetada para tratar os padrões lineares e não lineares das séries temporais, obtendo melhor desempenho quando comparadas aos seus componentes individuais. Por fim, o terceiro motivo relata um fato na literatura de previsão de séries temporais que devido ao comportamento das séries, a tarefa de mapear todos os padrões se torna difícil e estimula a pesquisa por novos sistemas de previsão.

Essa abordagem (duas fases) aplica o modelo linear diretamente nos dados, sem antes realizar um estudo sobre a natureza deles. Alguns modelos lineares como o ARIMA supõem linearidade e estacionaridade da série, portanto ao aplicá-los diretamente pode gerar um modelo mal especificado. Como consequência, o mapeamento dos padrões lineares pode não ser realizado totalmente, e o modelo não linear pode não ser capaz de mapeá-los adequadamente.

Para reduzir o problema da especificação de modelos, Babu e Reddy (BABU; REDDY, 2014) primeiro realizaram a decomposição da série temporal em termos de sua

natureza de distribuição. Foi utilizado um filtro MA para identificar padrões que seguem uma distribuição normal. Atingindo esse objetivo, a série restante é mapeada por uma RNA. Um abordagem similar foi proposta por Khandelwal *et al.* (KHANDELWAL; ADHIKARI; VERMA, 2015) onde a decomposição é realizada através de uma transformada discreta de wavelet, e o termo de baixa frequência e de alta frequência são tratados por modelos ARIMA e RNA, respectivamente.

Muitos trabalhos em previsão de séries temporais utilizam uma abordagem empírica para a seleção de parâmetros baseada em conhecimento do domínio ou em tentativa e erro (CRONE; KAUSCH; PREBMAR, 2004; ZHANG, 2003; BABU; REDDY, 2014; ZHANG; PATUWO; HU, 1998). Uma alternativa é utilizar algoritmos de otimização baseados em inteligência computacional para seleção de modelos (DONATE; CORTEZ, 2014; HSIEH; LEE; LEE, 2011; YOLCU; EGRIOGLU; ALADAG, 2013), apesar do custo computacional elevado, a abordagem automática pode produzir modelos mais precisos.

A abordagem de Box-Jenkins (BOX; JENKINS; REINSEL, 2013) faz uso de análises gráficas na fase de identificação do modelo ARIMA. Logo a identificação necessita de intervenção humana, podendo ocasionar erro de especificação. No entanto utilização de sistemas compostos por modelos sub-ajustados pode melhorar a acurácia das previsões (ZHANG, 2003).

Portanto, este trabalho tem como foco a utilização de algoritmos de otimização em sistemas que realizam a decomposição automática de séries temporais em termos de padrões lineares e não lineares.

1.2 OBJETIVOS

O presente trabalho tem por objetivo o desenvolvimento de novas metodologias para a previsão de séries temporais baseadas na decomposição das séries e utilizando algoritmos de otimização. A decomposição é realizada de forma automática, sem intervenção do especialista, por meio de um modelo de suavização exponencial. Variações do algoritmo PSO e o algoritmo genético (*genetic algorithm*) (GA) foram utilizadas tanto para a otimização das técnicas quanto para a decomposição da série. A decomposição da série é realizada em termos de sua distribuição de dados e utilizando a medida de curtose para determinar se a distribuição é Gaussiana (normal).

Alguns objetivos específicos desse trabalho são:

1. Explorar abordagens de decomposição de séries temporais em termos de padrões lineares e não lineares;
2. Realizar seleção automática de hiper-parâmetros em sistemas híbridos;
3. Analisar a natureza dos dados utilizando filtros baseados em suavização exponencial;

4. Analisar a influência de cada componente do sistema na previsão final;
5. Realização de experimentos para verificar o desempenho dos sistemas;

1.3 PRODUÇÃO BIBLIOGRÁFICA

Nesta seção, as produções bibliográficas referentes ao desenvolvimento deste trabalho são listadas. No total foram publicados três artigos em congressos, um artigo publicado em periódico (*Neurocomputing*) e um em processo de submissão.

1.3.1 Publicação em periódicos

- OLIVEIRA, J. F. de; LUDERMIR, T. B. A hybrid evolutionary decomposition system for time series forecasting. Neurocomputing, v.180, p.27 – 34, 2016.
- OLIVEIRA, J. F. de; LUDERMIR, T. B. Evolutionary semi-linear artificial neural network for time series forecasting., 2016. Submetido para periódico.

1.3.2 Publicação em congressos

- OLIVEIRA, J. F.; LUDERMIR, T. B. Iterative ARIMA-multiple support vector regression models for long term time series prediction. In: EUROPEAN SYMPOSIUM ON ARTIFICIAL NEURAL NETWORKS (ESANN), COMPUTATIONAL INTELLIGENCE AND MACHINE LEARNING., 2014.
- OLIVEIRA, J. F. Lorenzato de; B. LUDERMIR, T. A Hybrid Evolutionary System for Parameter Optimization and Lag Selection in Time Series Forecasting. In: INTELLIGENT SYSTEMS (BRACIS), BRAZILIAN CONFERENCE ON. 2014. p.73–78.
- OLIVEIRA, J. F. Lorenzato de; B. LUDERMIR, T. A distributed PSO-ARIMA-SVR hybrid system for time series forecasting. In: SYSTEMS, MAN AND CYBERNETICS (SMC), 2014 IEEE INTERNATIONAL CONFERENCE ON. 2014. p.3867–3872.

1.4 ORGANIZAÇÃO DO TRABALHO

Este trabalho está organizado em formato de artigos, no entanto alguns capítulos foram incluídos com o propósito de resumir o método proposto juntamente e experimentos adicionais. O Capítulo 2 resume o método proposto, o Capítulo 3 apresenta experimentos adicionais. Alguns gráficos das simuações adicionais estão disponíveis em http://goo.gl/6GNcID. O Capítulo 4 apresenta as conclusões do trabalho. Os conteúdos dos apêndices A ao C também serão discriminados. Os artigos abordam as estratégias que definiram a linha de pesquisa deste trabalho.

No apêndice A um algoritmo PSO com codificações discretas e contínuas é aplicado na seleção de modelos de um sistema ARIMA-SVR. A codificação binária é utilizada para encontrar a ordem do modelo ARIMA, e a parte contínua encontra os hiper-parâmetros do modelo SVR.

O apêndice B estende o modelo proposto no apêndice A aplicando seleção de características (*lags*) nas séries temporais. De acordo Zhang (ZHANG, 2001a) a dimensão das características possui uma grande influência no desempenho dos sistemas e a seleção de *lags* pode proporcionar um ganho de performance.

O apêndice C apresenta um sistema que utiliza um filtro de suavização exponencial para decompor a série em dados Gaussianos e não Gaussianos. O fator de suavização, a ordem do modelo autorregressivo e os hiper-parâmetros do modelo SVR são determinados por um algoritmo PSO. A curtose da série suavizada é utilizada para verificar sua natureza.

2 Método Proposto

A decomposição de séries temporais considerando padrões lineares e não lineares é uma estratégia para a previsão de séries temporais. O modelo linear ARIMA em diversos trabalhos é aplicado diretamente sobre os dados, o que pode ocasionar erro de especificação devido às suposições de linearidade e estacionaridade dos dados feitas pelo modelo.

O modelo MA utilizado por Babu e Reddy (BABU; REDDY, 2014) tem como objetivo extrair uma série linear de baixa volatilidade e uma série não linear de alta volatilidade, como mostrado na Figura 2. A ordem q do modelo MA é incrementada iterativamente até produzir uma série de baixa volatilidade. A série de baixa volatilidade é aplicada a um modelo ARIMA enquanto que a série de alta volatilidade é mapeada uma uma RNA.

Ao contrário da função de médias móveis onde é necessário armazenar as últimas q observações, além de atribuir o mesmo peso a todas elas, a suavização exponencial possui a vantagem de apenas armazenar a última observação, dando pesos gradativamente menores para observações passadas. O filtro de suavização exponencial é mostrado na equação (2.1) (MAKRIDAKIS; WHEELWRIGHT; HYNDMAN, 1998), onde s_t é a saída do modelo de suavização e α é o valor do coeficiente de suavização. A desvantagem dessa técnica é a necessidade de encontrar um valor adequado para α que pode variar conforme o problema aplicado.

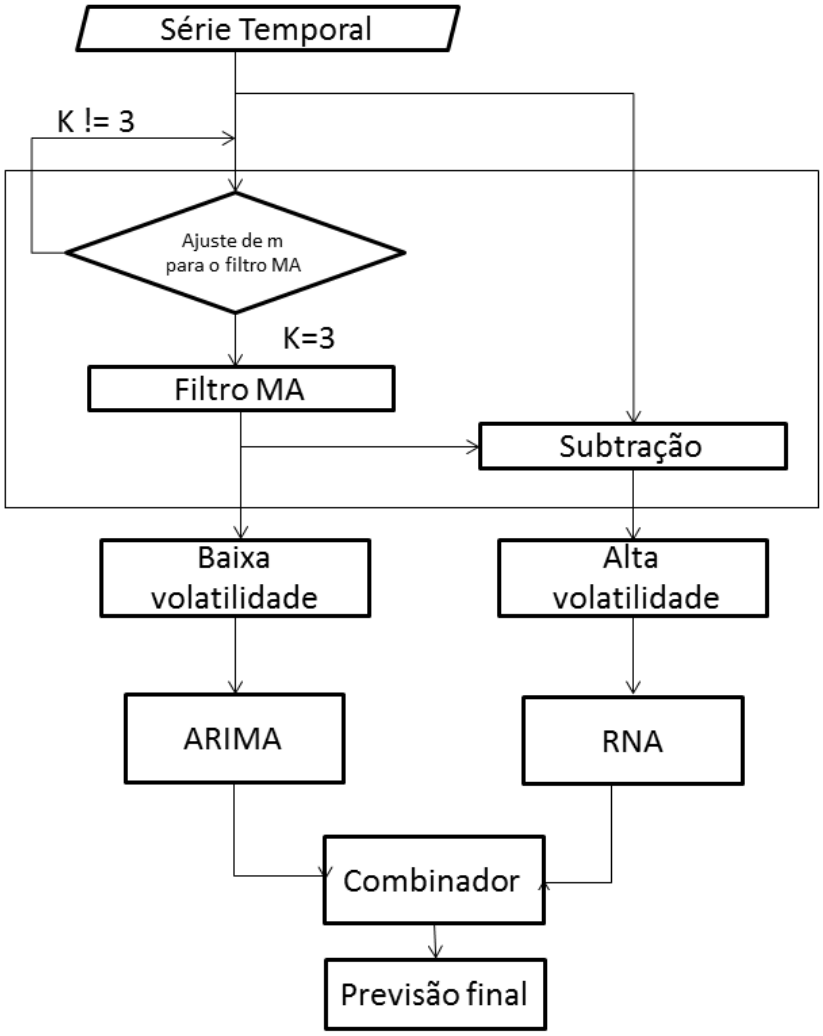
$$s_t = \alpha y_t + (1 - \alpha) s_{t-1}. \tag{2.1}$$

O filtro de suavização exponencial também realiza a divisão da série em termos Gaussianos e não Gaussianos. A componente não Gaussiana pode apresentar uma estrutura de dados composta por vários tipos de padrões (FAN; YAO, 2013) e pode ser considerada não linear (ROSENBLATT, 2000).

O método proposto é dividido em três fases: modelagem de padrões de baixa volatilidade (Gaussiano) e modelagem de alta volatilidade (não Gaussiano) que é realizada utilizando modelos lineares e não lineares. Os dados considerados de baixa volatilidade possuem propriedades similares à um processo Gaussiano com média 0 e variância constante, enquanto que os de alta volatilidade não possuem esta característica. Portanto, é assumido que o termo de baixa volatilidade, por possuir características Gaussianas, pode ser modelado por um processo ARIMA e o termo de alta volatilidade é mapeado por um sistema híbrido.

O primeiro passo é a aplicação da técnica de suavização exponencial para filtrar os dados de baixa volatilidade e os dados restantes (alta volatilidade) são modelados por um método híbrido, conforme a Figura 3. Nenhuma suposição de estacionariedade é feita

Figura 2 – Método de Babu e Reddy



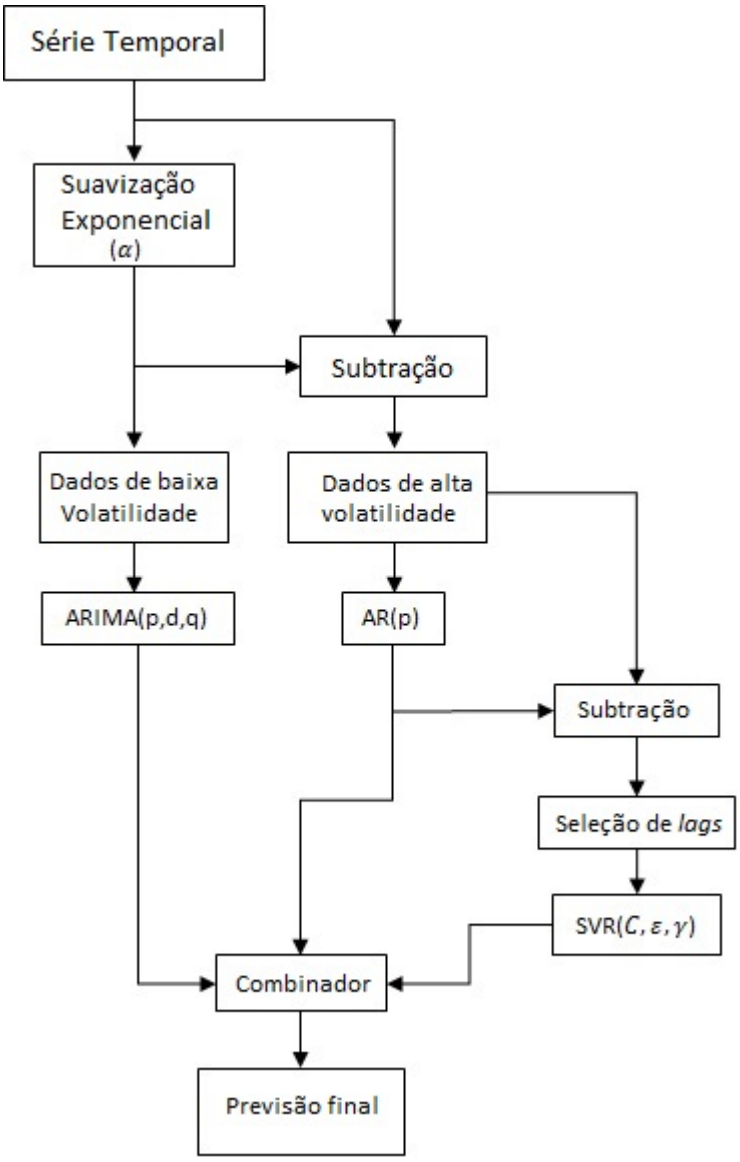
Fonte: (BABU; REDDY, 2014)

sobre a série inicial, e o modelo ARIMA é aplicado ao termo de baixa volatilidade.

A seleção dos parâmetros é realizada através de algoritmos de otimização. O PSO é explorado utilizando 5 variações: *Global* PSO (KENNEDY; EBERHART, 1995), *Constricted* PSO (CLERC; KENNEDY, 2002), *Modified* PSO (*time-varying inertia weight*) (SHI; EBERHART, 1998), PSO com topologia local (*lbest*) em anel (EBERHART; KENNEDY et al., 1995) e *Constricted* PSO com topologia local em anel. No mesmo contexto, um algoritmo genético (*Genetic Algorithm*) (GA) (EIBEN; SMITH, 2003) também é testado.

O PSO é foi escolhido devido à sua baixa complexidade computacional e por possuir uma memória de regiões promissoras passadas, fazendo com que o enxame tenha uma convergência nessas regiões. O GA por sua vez possui maior complexidade computacional devido aos operadores genéticos de seleção, cruzamento e mutação. O operador de cruzamento combina aspectos dos das soluções (pais), resultando em um novo indivíduo que pode não estar em uma região promissora. Essa característica é vantajosa para a

Figura 3 – Método Proposto



Fonte: do autor.

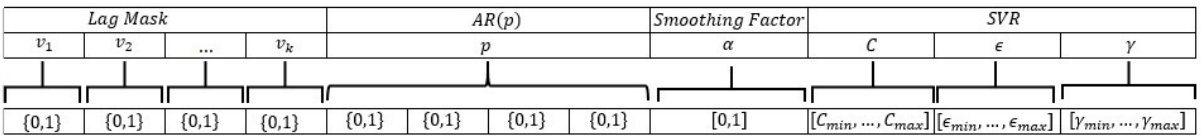
exploração do espaço de soluções.

2.1 ESPAÇO DE OTIMIZAÇÃO

A seleção de hiper-parâmetros do método proposto é realizada através de algoritmos de otimização. O espaço de busca foi definido com codificações discretas e contínuas como é mostrado na Figura 4.

O coeficiente de suavização α irá filtrar a série conforme a medida de curtose. De acordo com o teste de Jarque-Bera (JARQUE; BERA, 1980), um valor de curtose igual a 3 indica que os dados provém de uma distribuição gaussiana, logo poderiam ser modelados por um processo ARIMA. Valores de α próximos a 1 indicam praticamente nenhum tipo

Figura 4 – Espaço de otimização



Fonte: (OLIVEIRA; LUDERMIR, 2014)

de filtragem, ou seja, toda a série é passada para o modelo ARIMA, enquanto que valores próximos a 0 realizam uma filtragem mais intensa e nesse caso praticamente toda a série é passada para o módulo de mapeamento de dados de alta volatilidade.

Depois do processo de filtragem, o modelo AR é utilizado na primeira etapa do módulo de alta volatilidade. O modelo AR foi escolhido devido à sua simplicidade (CONNOR; MARTIN; ATLAS, 1994). Em trabalhos como o de Lapedes e Farber (LAPEDES; FARBER, 1987) RNAs são formuladas tomando como base modelos AR não lineares.

São empregados 4 bits para a codificação da ordem do modelo AR, que pode variar de 0 à 15, de acordo com a medida de erro utilizada em um conjunto de validação. Os resíduos produzidos pelo modelo AR são então atribuídos ao processo de elaboração do modelo SVR não linear.

Um importante parâmetro utilizado na construção de modelos é o número de observações passadas k (dimensão). De acordo com Takens (TAKENS, 1981) é possível modelar a série temporal empregando uma dimensão suficientemente larga.

O primeiro campo do espaço de otimização (Figura 4) corresponde à máscara de *lags* onde serão selecionados as observações passadas mais relevantes para a previsão do modelo SVR. Considere uma série $Y = [1, 3, 4, 10, 5, 6, 7]$, suponha o valor de $k = 3$, portanto a série redimensionada teria a seguinte estrutura $Y = [[1, 3, 4], [3, 4, 10], [4, 10, 5], [10, 5, 6]]$. Uma máscara de *lags* com configuração $[1, 0, 1]$ descartaria a segunda observação no vetor da série redimensionada, produzindo a seguinte série $Y = [[1, 4], [3, 10], [4, 5], [10, 6]]$. A função da máscara de lags é selecionar as observações passadas mais relevantes para o modelo a ser construído e possibilitar a redução da complexidade na etapa de elaboração do modelo SVR.

O modelo SVR utilizado nesta tese é constituído de uma função de kernel de base radial (OLIVEIRA; LUDERMIR, 2014; ADHIKARI, 2015) ($k(\mathbf{x}_i, \mathbf{x}_j) = exp(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\gamma^2})$), logo o parâmetro γ é utilizado. Para encontrar a melhor função f, é necessário minimizar uma função de risco estrutural (equação 2.2), onde $C > 0$ é um fator de regularização, $\|.\|$ é a norma do vetor e $L(.,.)$ é uma função de custo. O parâmetro C realiza um balanço entre a minimização dos pesos e a função de custo L mostrada na equação 2.3. O parâmetro ϵ atribui uma margem de erro, não contabilizando erros dentro dessa margem, logo o valor de ϵ não pode ser tão grande a ponto de não contabilizar nenhum erro, nem demasiadamente

pequeno podendo causar super ajustamento.

$$\frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{l} L(y_i, f(\mathbf{x}_i)) \tag{2.2}$$

Em problemas de regressão, encontra-se uma função $f(x)$ que produza saídas contínuas que desviem no máximo de ε do rótulo desejado.

$$L(y, f(\mathbf{x})) = \begin{cases} 0, & |f(\mathbf{x}) - y| < \varepsilon \\ |f(\mathbf{x}) - y| - \varepsilon, & \text{Caso contrário} \end{cases} \tag{2.3}$$

2.2 ALGORITMOS DE OTIMIZAÇÃO

A otimização de parâmetros é realizada através de variações do algoritmo baseado em enxame de partículas (PSO) e de um algoritmo genético (GA). A medida de performance de cada partícula ou cromossomo i é mostrada na equação (2.4).

$$fitness_i = (|3 - kurtosis(\mathbf{y})|) + \frac{1}{m}\sum_{j=1}^{m}(h_i - \hat{h}_i)^2. \tag{2.4}$$

Onde $\mathbf{y}$ representa a série produzida após o processo de suavização exponencial, $\hat{h}_t^i$ representa a i-ésima previsão do termo de alta volatilidade. Uma série Gaussiana possui valor de curtose igual a 3, portanto na minimização foi empregado o termo $(|3 - kurtosis(\mathbf{y})|)$, para que esse termo se aproxime de zero quando a curtose estiver próxima a 3. O termo seguinte busca reduzir o erro de previsão do modelo híbrido AR-SVR.

2.2.1 Particle swarm optimization

O PSO é um algoritmo de busca heurística inspirado em fenômenos da natureza (vôos de pássaros). Cada partícula representa uma possível solução para o problema e se move em direção à melhor solução encontrada pelo enxame (*gbest*) e à melhor solução individual (*pbest*). A velocidade das partículas é atualizda de acordo com a equação (2.5).

$$\mathbf{v}_i(t+1) = \mathbf{v}_i(t) + c_1 \cdot rnd() \cdot (\mathbf{pbest}_i - \mathbf{g}_i(t)) + c_2 \cdot rnd() \cdot (\mathbf{gbest} - \mathbf{g}_i(t)) \tag{2.5}$$

A adição de um fator de constrição χ (CLERC; KENNEDY, 2002), conforme a Equação (2.6), permite o melhor controle da velocidade, evitando assim possíveis estados de explosão que prejudicam a convergência do algoritmo.

$$\mathbf{v}_i(t+1) = \chi(\mathbf{v}_i(t) + c_1 \cdot rnd() \cdot (\mathbf{pbest}_i - \mathbf{g}_i(t)) + c2 \cdot rnd() \cdot (\mathbf{gbest} - \mathbf{g}_i(t))) \tag{2.6}$$

A regulagem entre exploração e explotação também influencia na performance do algoritmo. Shi e Eberhart (SHI; EBERHART, 1998) introduziram a noção de peso de inércia ω (Equação (2.7)) no processo de busca. Inicialmente, o ω assume o máximo valor estabelecido, incentivando a exploração do espaço de busca. Com o decorrer das iterações esse valor é reduzido, incentivando a busca em uma região específica desse espaço.

$$\mathbf{v}_i(t+1) = \omega(t)(\mathbf{v}_i(t) + c_1 \cdot rnd() \cdot (\mathbf{pbest}_i - \mathbf{g}_i(t)) + c2 \cdot rnd() \cdot (\mathbf{gbest} - \mathbf{g}_i(t))) \quad (2.7)$$

A abordagem global do PSO pode apresentar convergência prematura em algumas funções (ENGELBRECHT, 2013). Portanto a utilização de abordagens locais pode trazer benefícios para a otimização uma vez que a melhor solução pode ser definida através de uma vizinhança local.

A codificação binária é tratada utilizando o PSO binário (KENNEDY; EBERHART, 1997), onde é empregada uma transformação logística no valor da velocidade conforme é apresentado na equação (2.8)

$$\text{if rnd}() < Sig(\mathbf{v}_i), \text{ then } \mathbf{g}_i = 1; \text{else } \mathbf{g}_i = 0; \quad (2.8)$$

onde $Sig(\mathbf{v_i}) = \frac{1}{1+e^{-\mathbf{v_i}}}$.

2.2.2 Algoritmos Genéticos

Algoritmos genéticos são algoritmos de otimização que assimilam aspectos da evolução natural, como mutação, seleção e cruzamento. Os operadores de mutação e cruzamento são aplicados de acordo com as codificações no espaço de busca (EIBEN; SMITH, 2003). Nas codificações discretas que compõem parte do espaço de busca o operador de mutação é aplicado por meio de troca de bits, enquanto que o cruzamento de dois pontos é aplicado na codificação dos *lags* e da ordem do modelo AR. Nas codificações contínuas a mutação não uniforme com distribuição normal é aplicada e o cruzamento é feito por recombinação artimética como mostrado nas Equações 2.9 e 2.10, onde η é o parâmetro utilizado no operador que fica limitado no intervalo $[0, 1]$.

$$Child_1 = \eta \cdot Parent_1 + (1-\eta) \cdot Parent_2 \quad (2.9)$$

$$Child_2 = \eta \cdot Parent_2 + (1-\eta) \cdot Parent_1 \quad (2.10)$$

3 Experimentos Adicionais

Os experimentos adicionais foram realizados com o propósito de analisar o desempenho do sistema proposto utilizando versões aprimoradas do PSO que abrangem regulação das habilidades de exploração e explotação, controle de velocidade e utilização de topologias locais. Também, uma análise residual é realizada para verificar a importância de cada componente do sistema híbrido. São utilizados: *constricted* PSO (CLERC; KENNEDY, 2002), PSO com variação dos pesos de inércia *V. Intertia* (SHI; EBERHART, 1998), *constricted* PSO com topologia local em anel (*constricted R.*), PSO global, PSO com topologia local em anel (*Ring*) e um algoritmo genético (GA).

Tabela 1 – Parâmetros dos algoritmos de otimização

Method	w	w_{min}	χ	CR	MR	η
Constricted PSO	-	-	0.7298	-	-	-
V. Inertia	0.9	0.4	-	-	-	-
Constricted R.	-	-	0.7298	-	-	-
Global	1	-	-	-	-	-
Ring	1	-	-	-	-	-
GA	-	-	-	0.75	0.1	0.75

O número de *lags* k foi determinado através de análise da função de autocorrelação das séries. Foram realizados experimentos utilizando 100 e 200 iterações nos algoritmos de otimização, no entanto não foi observado nenhuma redução de erro expressiva após 100 iterações nas bases testadas. Portanto, o PSO e o GA são executados por 100 iterações e o tamanho da população foi fixado em 30. O número de iterações e o tamanho da população foram definidos com base em estudos presentes na literatura (SELAKOV et al., 2014; HSIEH; LEE; LEE, 2011; HUANG; DUN, 2008). Os coeficientes pessoal e social c_1 e c_2 foram fixados em 2 em todas as variações do PSO com exceção nas abordagens baseadas em *Constricted* PSO que assumem valores $c_1 = 2.05$ e $c_2 = 2.05$. Na tabela 1 são mostrados os demais parâmetros, onde CR e MR correspondem às taxas de cruzamento e mutação, respectivamente. No algoritmo genético um operador de elitismo é empregado, aproveitando 50% do melhores indivíduos da geração passada para a geração atual. A seleção proporcional ao fitness é empregada para a seleção de indivíduos.

3.1 ANÁLISE DE DESEMPENHO

O experimento foi repetido 30 vezes e o erro quadrático médio (*mean square error*) (MSE) foi utilizado como medida de análise de desempenho no conjunto de testes. As bases de dados são oriundas de problemas reais (HYNDMAN, 2010) e são exploradas em vários trabalhos na literatura (CORTEZ, 2010; ZHANG, 2003; BABU; REDDY, 2014;

OLIVEIRA; LUDERMIR, 2016). Os valores médios e desvios padrão são apresentados na Tabela 2.

Tabela 2 – Valores médios e desvios padrão (entre parêntesis) do MSE

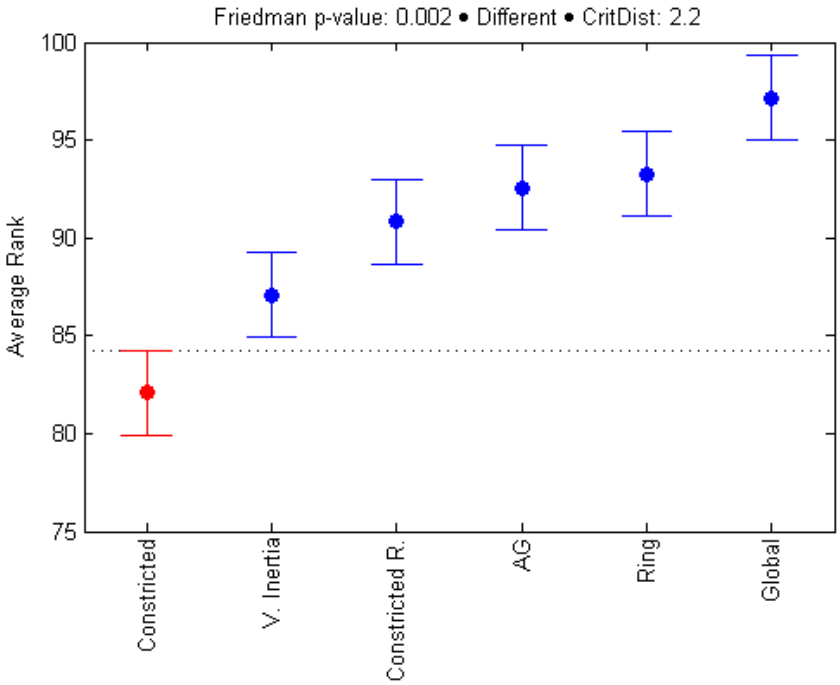
Dataset	Constricted	V. Inertia	Constricted R.	Global	Ring	GA
Dowjones	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005
	(2.67e-10)	(1.47e-10)	(3.66e-10)	(1.48e-10)	(1.42e-10)	(1.54e-10)
Accidental Death	0.0035	0.0033	0.0035	0.0030	0.0029	0.0040
	(0.0010)	(0.0010)	(0.0016)	(0.0007)	(0.0002)	(0.0017)
Electricity	0.0014	0.0011	0.0011	0.0020	0.0016	0.0017
	(0.0010)	(6.4121e-05)	(4.6185e-05)	(0.0009)	(0.0007)	(0.0012)
IBM	0.0006	0.0006	0.0006	0.0006	0.0006	0.0006
	(8.27e-05)	(8.50e-05)	(0.0002)	(5.65e-05)	(6.28e-05)	(6.87e-05)
Lake Erie	0.0012	0.0012	0.0012	0.0012	0.0012	0.0012
	(8.03e-10)	(8.56e-10)	(6.04e-10)	(1.03e-09)	1(.14e-09)	(5.90e-10)
lynx	0.0073	0.0071	0.0072	0.0066	0.0113	0.0067
	(0.0021)	(0.0011)	(0.0011)	(0.0007)	(0.0259)	(0.0004)
NSW	0.0017	0.0019	0.0020	0.0020	0.0019	0.0021
	(0.0007)	(0.0010)	(0.0015)	(0.0012)	(0.0017)	(0.0020)
Pollution	0.0150	0.0144	0.0152	0.0152	0.0146	0.0153
	(0.0013)	(0.0005)	(0.0023)	(0.0015)	(0.0013)	(0.0029)
Colorado River	0.0036	0.0035	0.0037	0.0039	0.0037	0.0034
	(0.0011)	(0.0006)	(0.0014)	(0.0010)	(0.0011)	(0.0001)
Stock	0.0073	0.0073	0.0073	0.0073	0.0073	0.0073
	(6.38e-09)	(4.24e-09)	(4.01e-09)	(4.46e-09)	(3.93e-09)	(5.79e-09)
Sunspot	0.0100	0.0103	0.0106	0.0100	0.0099	0.0096
	(0.0008)	(0.0006)	(0.0008)	(0.0006)	(0.0005)	(0.0007)
Passengers	0.0012	0.0012	0.0013	0.0018	0.0016	0.0014
	(0.0001)	(7.70e-05)	(0.0004)	(0.0020)	(0.0007)	(0.0001)
Wine	0.0216	0.0215	0.0186	0.0226	0.0230	0.0214
	(0.0036)	(0.0038)	(0.0020)	(0.0019)	(0.0019)	(0.004)

Quando vários modelos são comparados, o número de comparações é maior, portanto um teste adequado deve ser empregado. O teste de *Friedman* (DEMŠAR, 2006) realiza ranqueamentos para cada técnica em cada base. Os ranques médios são apresentados na Tabela 3. No entanto, o teste de Friedman aponta que existe uma diferença entre os desempenhos dos algoritmos utilizados mas não indica quais algoritmos possuem essa diferença. Para conseguir essa informação, pode ser aplicado um pós-teste. No teste *post-hoc* de Nemenyi (DEMŠAR, 2006) o desempenho das técnicas é significativamente distinto se a diferença de ranques médios for maior que a distância crítica calculada utilizando uma

Tabela 3 – Ranking Médio obtido.

Variação	Ranking Médio
Constricted	83.2397
V. Inertia	86.9590
Constricted R.	89.7692
Global	97.1256
Ring	92.9769
GA	92.9295

Figura 5 – Resultado do teste de hipótese de Friedman-Nemenyi



Fonte: do autor

confiança de 95%.

O melhor desempenho foi obtido pela otimização utilizando a variação *constricted* PSO (CLERC; KENNEDY, 2002) conforme mostrado pelo resultado obtido através de um teste de *Friedman- Nemenyi* na Figura 5. Com um valor de $p - value = 0.002$ é possível rejeitar a hiótese nula de que os algoritmos possuem desempenho semelhante em termos de MSE. Logo para realizar comparações aos pares, a distância crítica calculada (2.2) é utilizada como parâmetros para verificar quais modelos são significativamente distintos dos outros, com base nos ranques médios. Os gráficos de previsões médias são apresentados na Figura 7.

Além de reduzir o erro de previsão, um objetivo esperado do sistema proposto é que seja capaz de extrair uma série Gaussiana com valor de curtose igual a 3 e outra série não Gaussiana com curtose diferente de 3. Em quatro bases de dados mostradas na Figura 6 esse objetivo não foi alcançado (Dow Jones, Stock, IBM e Lake Erie). As bases Dow Jones e Stock, possuem mudanças de nível ocasionadas por uma tendência exponencial. A base IBM possui uma mudança de nível na sua distribuição e a base Lake Erie possui além das variações sazonais, possui variações de nível.

A suavização exponencial simples utilizada supõe nível constante nas séries temporais, portanto não demonstrou capacidade para mapear os padrões necessários nesses casos apresentados. O valor do fator de suavização α ficou próximo a 1 levando praticamente toda

Tabela 4 – Melhor desempenho em termos de MSE

Dataset	Constricted	V. Inertia	Constricted R.	Global	Ring	GA
Dowjones	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005
Accidental Death	0.0024	0.0021	0.0024	0.0019	0.0027	0.0027
Electricity	0.0012	0.0010	0.0010	0.0015	0.0013	0.0014
IBM	0.0005	0.0005	0.0005	0.0005	0.0005	0.0005
Lake Erie	0.0012	0.0012	0.0012	0.0012	0.0012	0.0012
lynx	0.0056	0.0058	0.0058	0.0058	0.0056	0.0058
NSW	0.0016	0.0017	0.0016	0.0018	0.0017	0.0019
Pollution	0.0134	0.0130	0.0129	0.0137	0.0134	0.0133
Colorado River	0.0023	0.0026	0.0022	0.0024	0.0025	0.0030
Stock	0.0073	0.0073	0.0073	0.0073	0.0073	0.0073
Sunspot	0.0097	0.0097	0.0099	0.0101	0.0100	0.0098
Passengers	0.0011	0.0011	0.0012	0.0012	0.0013	0.0013
Wine	0.0168	0.0176	0.0153	0.0176	0.0217	0.0169

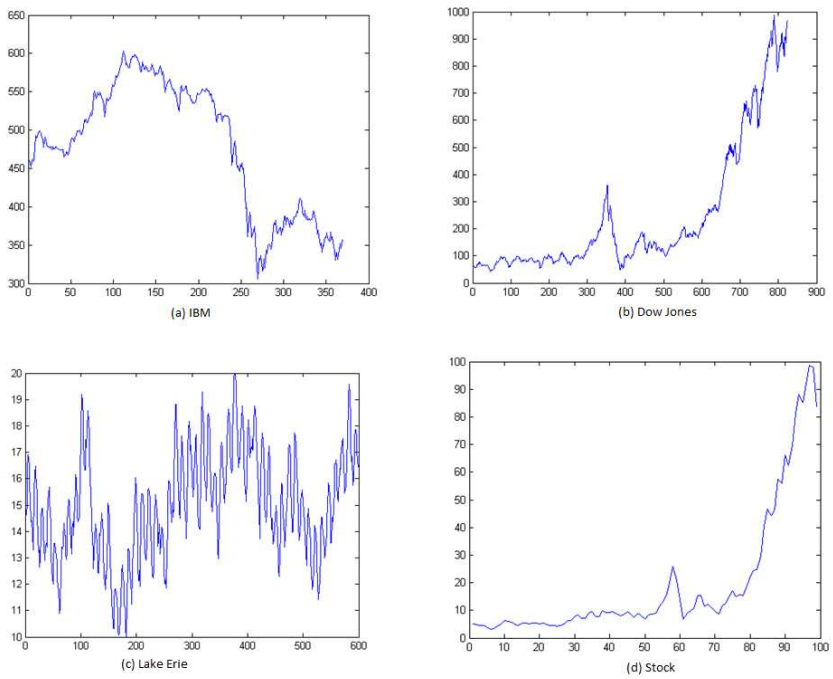


Figura 6 – Bases onde o sistema não obteve o desempenho esperado.

a série temporal ser mapeada pelo modelo ARIMA. Pequenas flutuações são mapeadas pelos modelos AR-SVR.

As séries NSW, Colorado River, Sunspot, Accidental Death e Lynx, possuem nível constante, além de possuírem padrões cíclicos. Na base Lynx, uma prática comum é aplicar uma operação logarítmica nos valores da série (ZHANG, 2003; KHASHEI; BIJARI, 2011), neste trabalho não foi aplicado o logaritmo, gerando previsões imprecisas como pode ser observado na Figura 7.

As bases Passengers, Electricity, Wine e Pollution possuem tendências que alteram o nível da série, no entanto essas as tendências são removidas e adicionadas ao termo de alta volatilidade. As bases Passengers e Electricity obtiveram bons resultados.

As bases NSW, Colorado River, Wine e Pollution apresentam variâncias não constantes, podendo dificultar a previsão.

A convergência dos dados é mostrada na Figura 8 onde é analisada a convergência média. O método baseado no *Constricted* PSO foi analisado individualmente na Figura 9.

Os métodos baseados em variação do peso de inércia (*V. Inertia*) (SHI; EBERHART, 1998) e *Constricted* PSO obtiveram as melhores convergências.

3.2 ANÁLISE RESIDUAL

Os resíduos produzidos pelo modelo proposto foram analisados para verificar se existem correlações pendentes. Para este fim, foi aplicada a função de autocorrelação nos resíduos juntamente com o expoente de *Hurst* (HURST R. P. BLACK, 1966) que pode ser usado como uma medida de previsibilidade das séries temporais (RASHEED; QIAN, 2004). O expoente de *Hurst* assume valores no inervalo [0...1], onde valores próximos a 0.5 indicam que a previsibilidade se torna mais difícil.

Entretanto de acordo com Granero *et al.* (GRANERO; SEGOVIA; PéREZ, 2008) a estimativa do expoente de *Hurst* via análise R/S exibe evidências de previsibilidade em séries aleatórias quando a série não é longa o suficiente. Além do mais, um parâmetro utilizado pela abordagem R/S é o tamanho mínimo da sub-série m, que também influencia no resultado.

A análise de *Hurst* conduzida leva em consideração um tamanho mínimo de subsérie m igual ao menor divisor da série temporal. Os resíduos produzidos em cada etapa do sistema híbrido sao analisados, portanto já que o módulo AR e SVR são sequenciais, é esperado que haja uma aproximação a 0.5 para evidenciar a influência do SVR. O resíduo produzido pelo modelo ARIMA corresponde à série Gaussiana que é fornecida e o resíduo produzido pela soma das previsões também é analisado.

A técnica baseada no *constricted PSO* obteve melhores resultados e foi escolhida para a análise. Primeiramente o resíduo médio obtido nas 30 execuções dos experimentos é analisado e os resultados obtidos pelo expoente de hurst são mostrados na Tabela 5. Os resíduos da melhor execução do experimento (melhor resultado em termos de MSE) também são analisados na Tabela 6.

De acordo com os resultados da Tabela 5, durante a execução do módulo composto pelas técnicas AR-SVR, o valor do expoente de hurst se aproxima de 0.5 em 10 das 13 bases, o que mostra a influência do SVR no mapeamento de resíduos produzidos pelo modelo AR. Na Tabela 6 os resíduos produzidos pela melhor execução são analisados. O valor do coeficiente de *Hurst* se aproximou de 0.5 em 7 das 13 bases.

A análise do expoente de Hurst utilizando os parâmetros citados nas séries se de-

Tabela 5 – Expoente de *Hurst* aplicado aos resíduos médios

Dataset	AR	SVR	ARIMA	Final
Dowjones	0.5468	0.5257	0.5201	0.5262
Accidental Death	0.4094	0.4709	0.4451	0.4759
Electricity	0.4234	0.4493	0.5933	0.4483
IBM	0.5172	0.5372	0.5033	0.5343
Lake Erie	0.5148	0.5185	0.4669	0.4986
Lynx	0.4706	0.4347	0.4814	0.4348
NSW	0.5366	0.5403	0.4985	0.5402
Pollution	0.4779	0.4956	0.5402	0.4825
Colorado River	0.4806	0.4869	0.5060	0.4937
Stock	0.4832	0.4926	0.4318	0.4761
Sunspot	0.5329	0.5498	0.5091	0.5461
Wine	0.3709	0.3768	0.4638	0.4032
Passengers	0.4306	0.4514	0.4509	0.4424

monstrou inconclusiva, pois houve variação nos resultados. McCauley *et al.* (MCCAULEY; GUNARATNE; BASSLER, 2007) analisa o expoente de *Hurst* e conclui que sua utilização pode ser insuficiente para obter informações sobre a dinâmica do processo e o recomendado seria realizar uma análise de autocorrelação.

As funções de autocorrelação para as bases Accidental Death, Colorado River, Wine e Electricity são apresentadas nas Figuras 10, 11, 12, 13. Para o cálculo dos intervalos de confiança (95% de confiança), os resíduos são considerados como ruído branco, logo o intervalo é computado através da Equação 3.1, onde N é o tamanho da amostra utilizada para o cálculo (BOX; JENKINS; REINSEL, 2013).

$$Intervalo = \frac{2}{\sqrt{N}} \tag{3.1}$$

As funções de autocorrelação foram aplicadas nos resíduos de cada módulo (AR, SVR e ARIMA) do sistema proposto e também no resíduo produzido pela combinação de resultados.

Os resíduos obtidos na base *accidental death* (Figura 10), indicam que o SVR teve pouca influência na redução de correlações. O fator de suavização obtido foi $\alpha = 0.3181$ em todas as simulações, indicando que tanto o ARIMA quanto o sistema composto pelo AR-SVR influenciaram o resultado final. A série não Gaussiana produzida possui características de estacionaridade evidenciada pelo teste KPSS (KWIATKOWSKI et al., 1992), portanto o modelo AR foi suficiente para mapear os padrões da série.

O fator de suavização obtido na simulação da base *Colorado River* (Figura 11) foi $\alpha = 0.0952$, indicando que o modelo ARIMA teve pouca participação nos resultados portanto, o módulo AR-SVR foi predominante na previsão. Observa-se também a redução

Tabela 6 – Expoente de *Hurst* aplicado aos resíduos da melhor execução

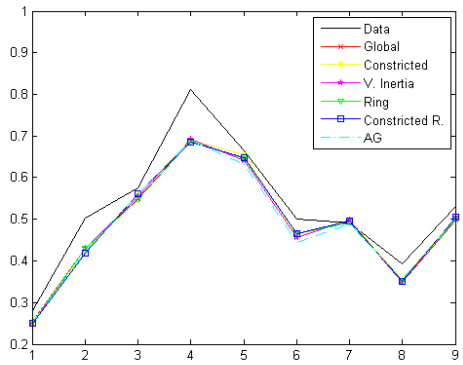
Dataset	AR	SVR	ARIMA	Final
Dowjones	0.5782	0.5265	0.5201	0.5259
Accidental Death	0.4079	0.4846	0.4451	0.4694
Electricity	0.4256	0.4815	0.5151	0.4807
IBM	0.5135	0.5069	0.5451	0.5070
Lake Erie	0.5267	0.5289	0.4669	0.5142
Lynx	0.4751	0.4762	0.4764	0.4614
NSW	0.5243	0.5633	0.5289	0.5633
Pollution	0.4602	0.4462	0.5402	0.4499
Colorado River	0.4708	0.4650	0.5060	0.4704
Stock	0.4979	0.5204	0.4318	0.4574
Sunspot	0.4881	0.5052	0.5084	0.5098
Wine	0.6038	0.3664	0.4638	0.3482
Passenger	0.4353	0.4794	0.4509	0.4743

de correlações do modelo AR para o modelo SVR.

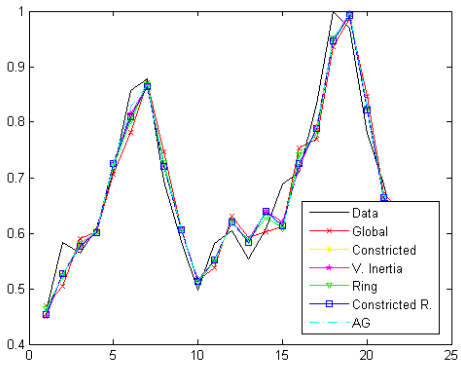
Na base *wine* (Figura 12), o coeficiente de suavização obtido foi de $\alpha = 0.5216$ atribuindo a mesma importância para os dois módulos. As correlações também foram reduzidas do modelo AR para o modelo SVR.

Em relação a base *electricity* (Figura 13), na maioria das execuções o valor do coeficiente foi $\alpha = 0,0400$, ocasionando pouca influência do modelo ARIMA.

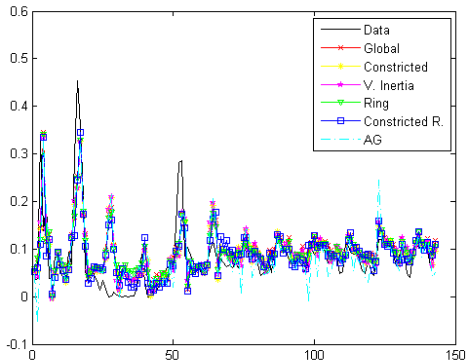
Figura 7 – Gráficos do conjunto de testes



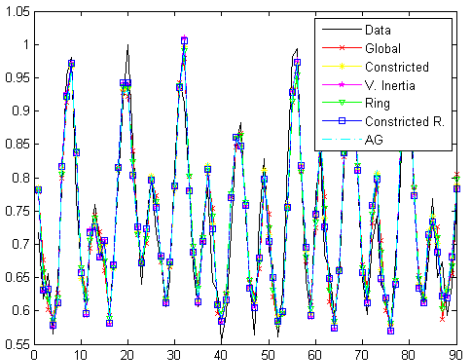
(a) Accidental Death



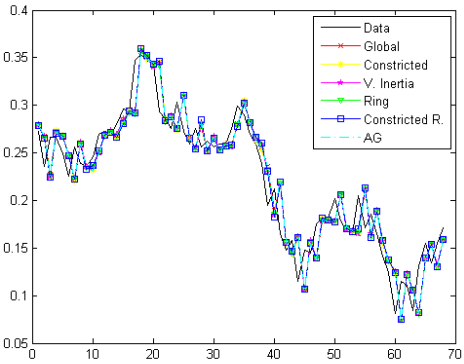
(b) Passengers



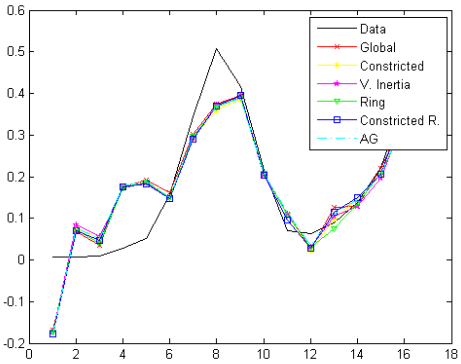
(c) Colorado River



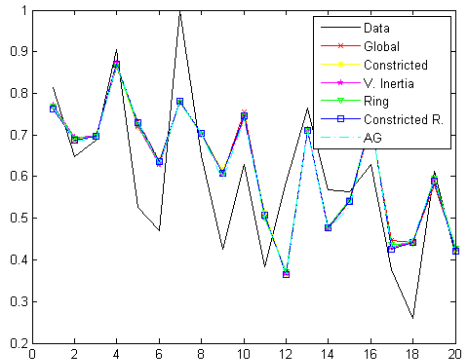
(d) Electricity



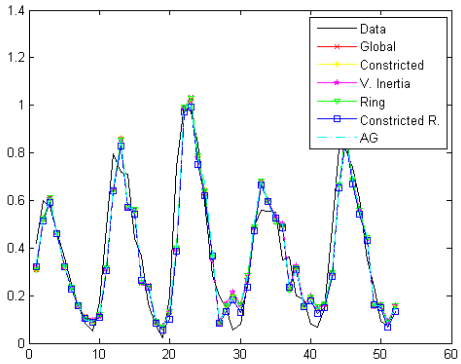
(e) IBM



(f) Lynx

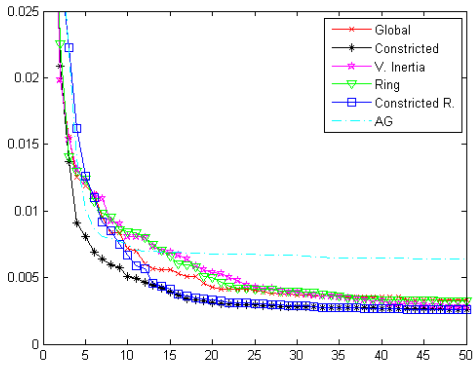


(g) Pollution

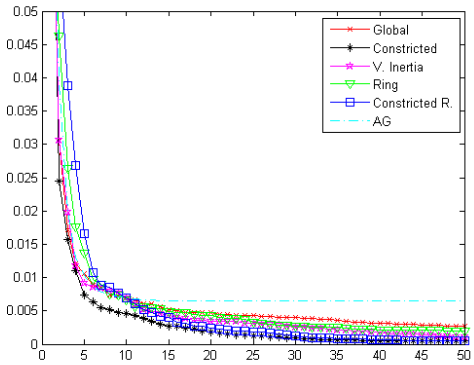


(h) Sunspot

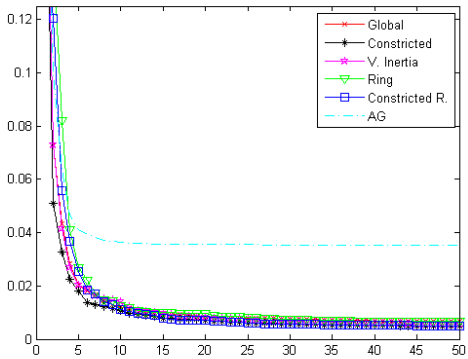
Figura 8 – Convergência dos métodos



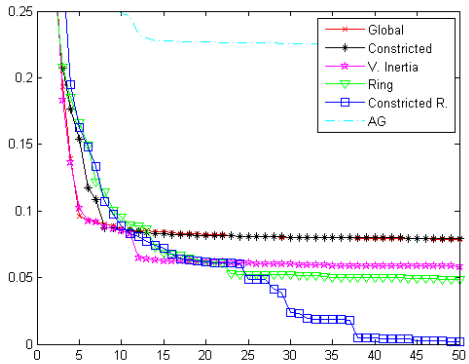
(a) Accidental Death



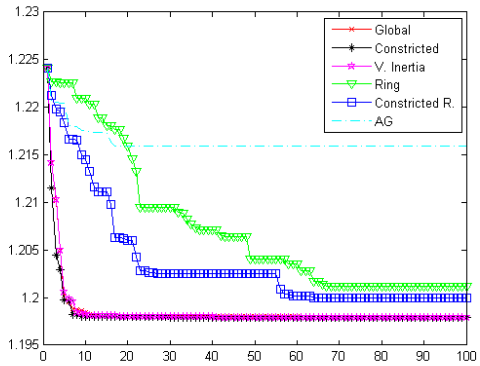
(b) Passengers



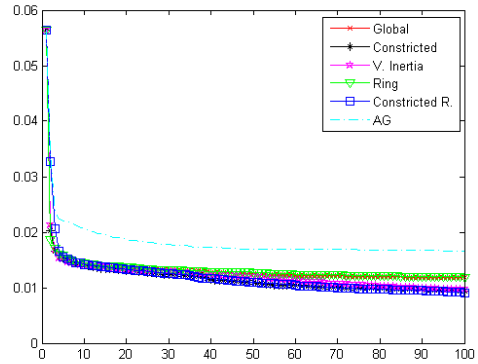
(c) Colorado River



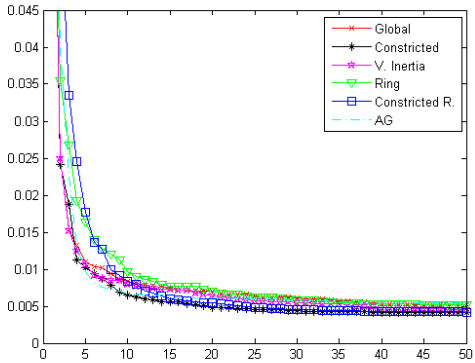
(d) Electricity



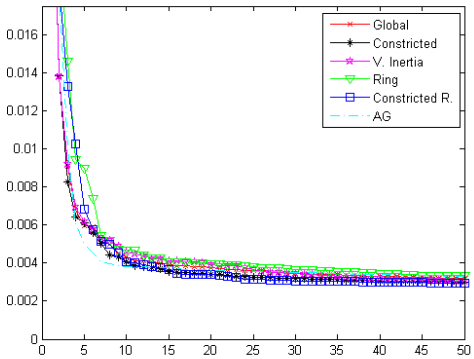
(e) IBM



(f) Lynx

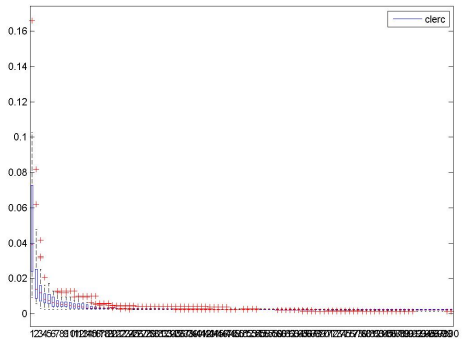


(g) Pollution

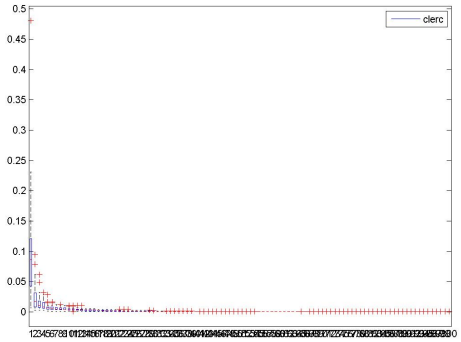


(h) Sunspot

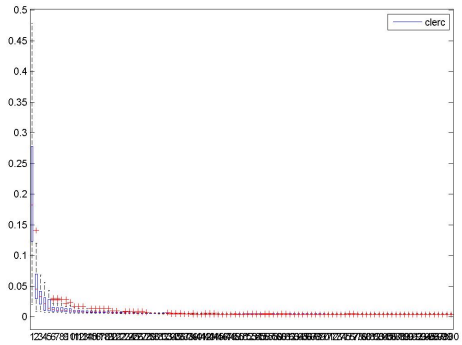
Figura 9 – Convergência do método *Constricted* PSO



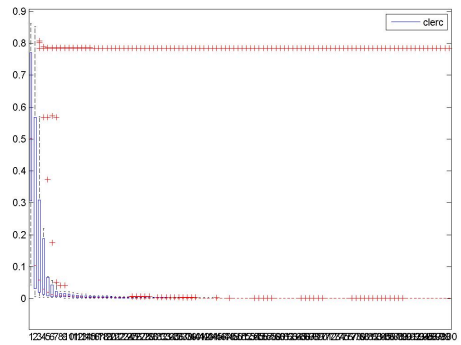
(a) Accidental Death



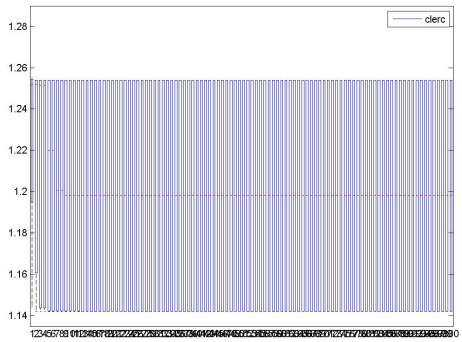
(b) Passengers



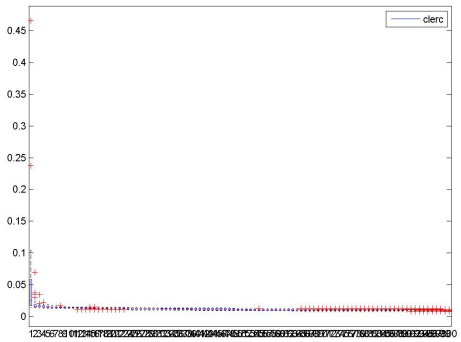
(c) Colorado River



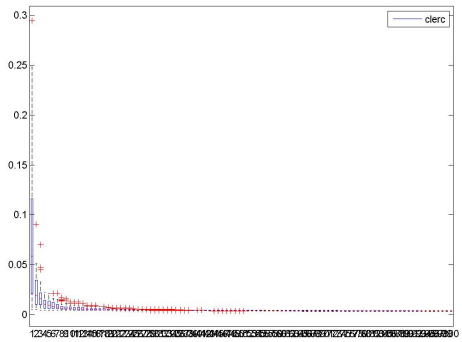
(d) Electricity



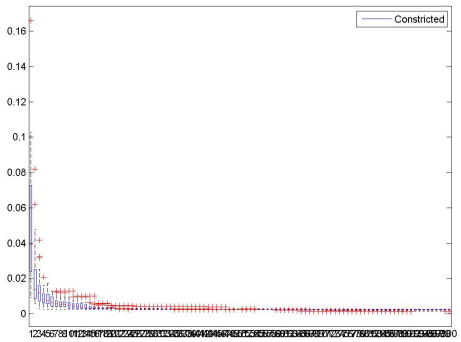
(e) IBM



(f) Lynx

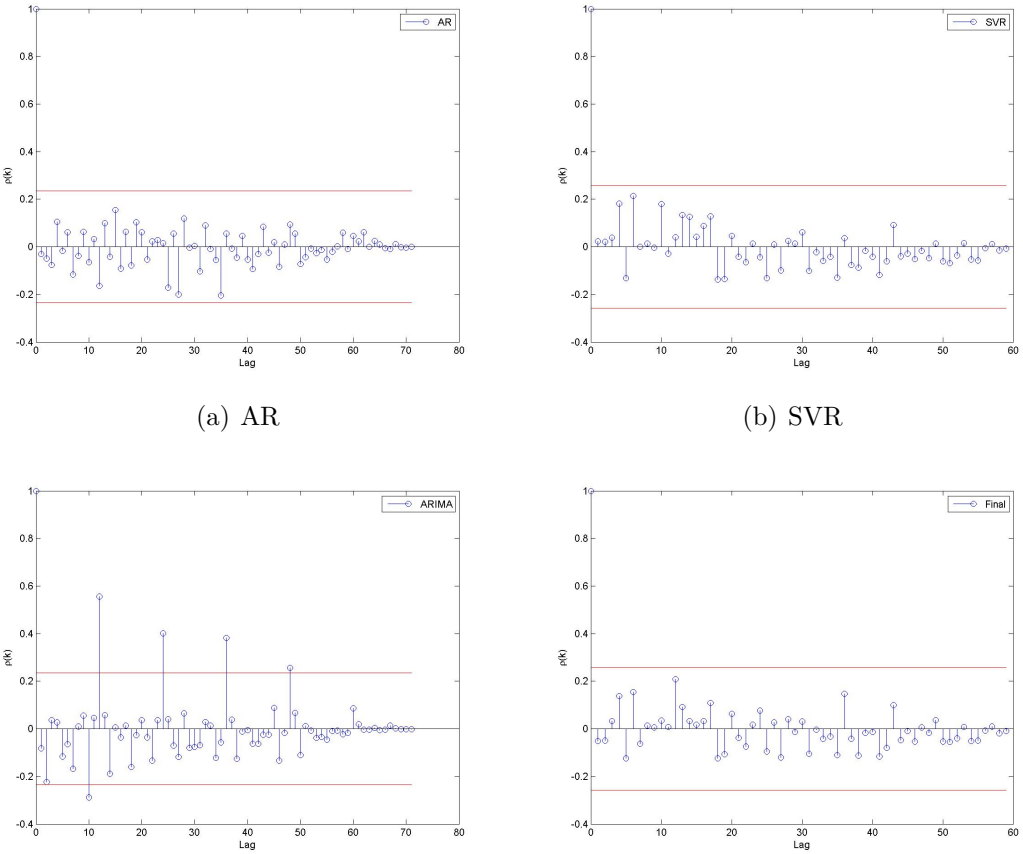


(g) Pollution



(h) Sunspot

Figura 10 – Análise de resíduo médio para base Accidental Death



(a) AR

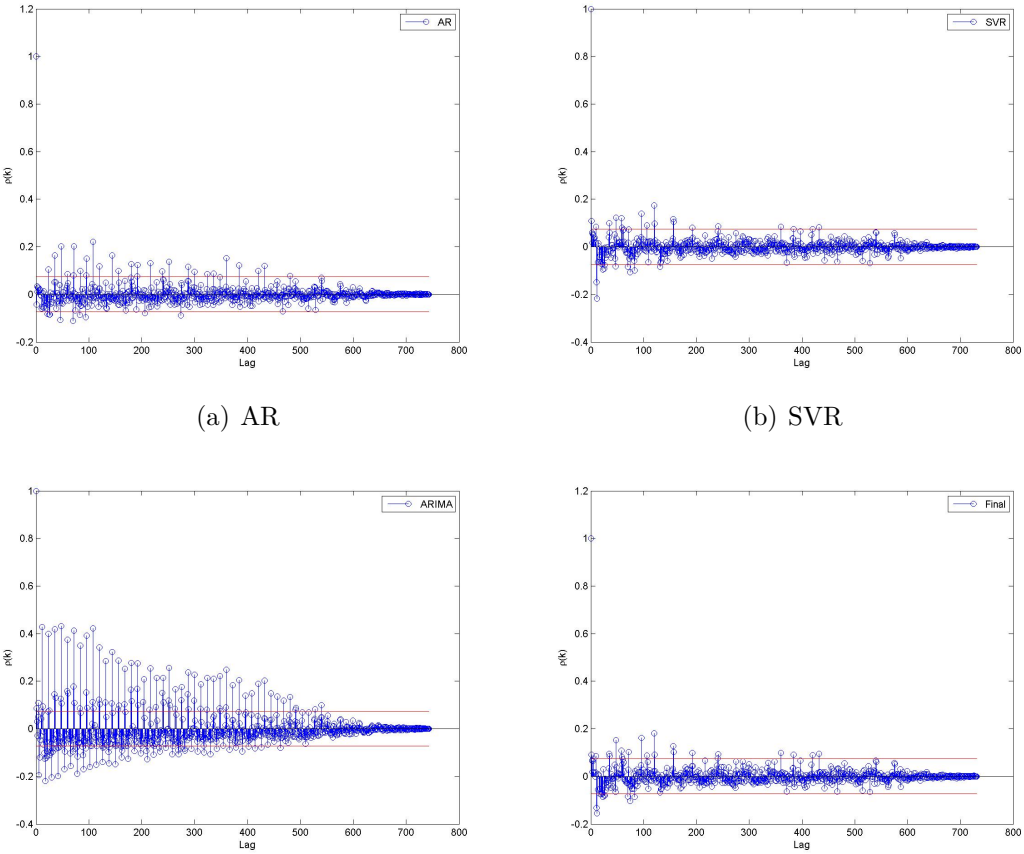
(b) SVR

(c) ARIMA

(d) Final

Fonte: do autor

Figura 11 – Análise de resíduo médio para base Colorado River



(a) AR

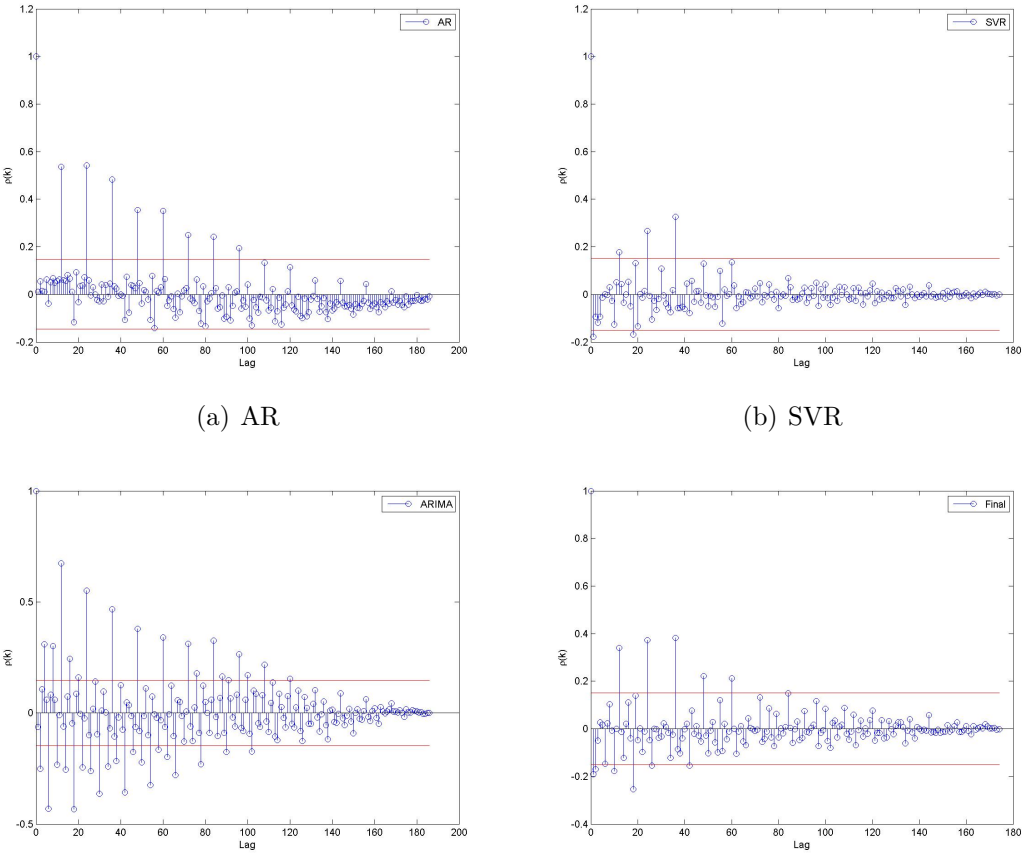
(b) SVR

(c) ARIMA

(d) Final

Fonte: do autor

Figura 12 – Análise de resíduo médio para base Wine



(a) AR

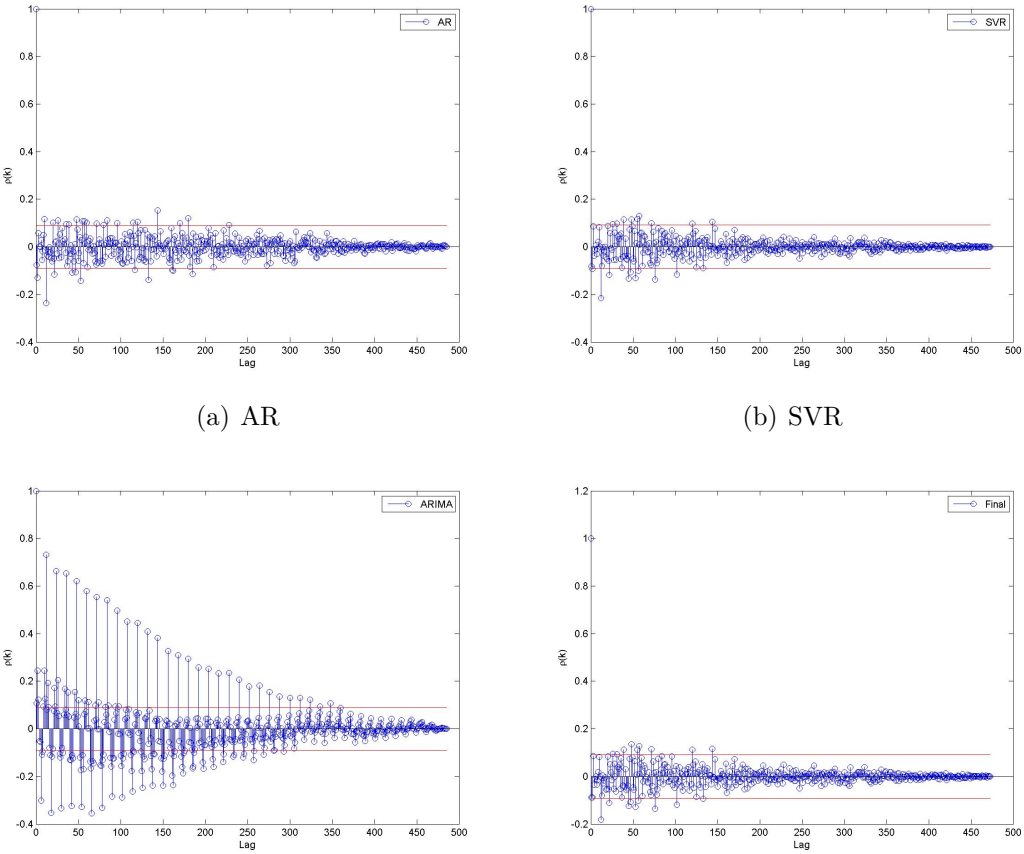
(b) SVR

(c) ARIMA

(d) Final

Fonte: do autor

Figura 13 – Análise de resíduo médio para base Eletricity



(a) AR

(b) SVR

(c) ARIMA

(d) Final

Fonte: do autor

4 Conclusões

Este trabalho apresenta uma abordagem de sistemas evolucionários baseados em decomposição aplicados à previsão de séries temporais. Os sistemas são compostos por modelos lineares e não-lineares para realizar o mapeamento dos respectivos padrões. Em relação ao estado-da-arte, o método proposto se destaca por utilizar uma abordagem evolucionária para decomposição das séries temporais por meio de um fitro de suavização exponencial simples e para otimização de hiper-parâmetros das técnicas de previsão.

As técnicas com otimização dos parâmetros apresentaram melhores resultados (apêndices A e B) em comparação com as técnicas híbridas tradicionais. A otimização dos hiper-parâmetros (ordem do modelo ARIMA, e os hiperparâmetros do SVR) foi realizada por um GA e por variações de PSO utilizando codificações discretas e contínuas. Os experimentos foram realizados em bases de problemas reais (HYNDMAN, 2010). Em especial, a seleção não uniforme de *lags* apresentada no apêndice B obteve melhorias em seus resultados.

No apêndice C foi utilizado um filtro de suavização exponencial para estudar a natureza da série. A série normal (curtose = 3) é tratada por um modelo ARIMA e a série restante é tratada por um sistema AR-SVR. Uma distribuição não-Gaussiana pode apresentar dados oriundos de distribuições lineares ou não-lineares (FAN; YAO, 2013), portanto foi utilizada uma técnica híbrida para realizar sua modelagem. Os resultados obtidos foram promissores, no entanto o sistema apresentou alta complexidade computacional.

Os experimentos adicionais contemplam a utilização de diferentes algoritmos de otimização para análise da metodologia proposta e também a análise residual. O sistema proposto apresentou melhoria nos resultados utilizando variações do algoritmo PSO, em particular os algoritmos de otimização baseados em fatores de constrição (CLERC; KENNEDY, 2002) apresentaram os melhores resultados em relação aos demais algoritmos, devido às suas capacidades de busca global e local. A otimização por algoritmos genéticos apresentou convergência prematura em algumas bases, possivelmente pela utilização de um operador de seleção proporcional ao *fitness* (EIBEN; SMITH, 2003).

A suavização exponencial simples supõe que a série temporal apresente nível constante, ou seja não possua padrões de tendência que modifique seu nível. Séries que apresentaram tendências exponenciais, polinomiais ou mudanças de nível (Dow Jones, Stock, Lake Erie e IBM) ocasionaram queda de desempenho, pois não foi possível o mapeamento correto dos padrões para a extração de uma série Gaussiana.

O expoente de *Hurst* foi utilizado para analizar a previsibilidade dos resíduos e a contribuição do modelo SVR no sistema. Portanto uma aproximação em relação a

0.5 do modelo AR para o modelo SVR, indicaria que o modelo SVR obteve êxito nas previsões, deixando a série menos previsível. Esse comportamento foi observado em 10 bases, em relação ao resíduo médio obtido. No entanto sua utilização pode apresentar resultados variados em decorrência do tamanho da série e do tamanho mínimo da sub-série (parâmetro utilizado para a estimativa do coeficiente de hurst) (GRANERO; SEGOVIA; PéREZ, 2008).

O expoente de *Hurst* também não seria indicado como metodologia única para extrair informações sobre a dinâmica do processo (MCCAULEY; GUNARATNE; BASSLER, 2007). Portanto, também foi realizada uma análise residual por meio de funções de autocorrelação. Foi observado a redução de correlações entre um resíduos no sistema AR-SVR em algumas bases, indicando a importância do modelo SVR no sistema.

Em geral, apesar do sistema proposto apresentar alto custo computacional, se mostrou eficiente em séries temporais com diferentes características. No entanto é necessário remover dos padrões de tendência da série como pré-processamento, devido à limitação imposta pelo filtro de suavização exponencial simples.

4.1 TRABALHOS FUTUROS

Durante a pesquisa foram identificados diversos temas que poderão ser abordados no futuro. As seguintes linhas se destacam para trabalhos a serem desenvolvidos após a conclusão do doutorado:

- **Métodos de combinação**: Durante esse trabalho, foi assumida uma relação linear entre a previsão do modelo linear e a previsão do modelo não-linear, sendo adotado um modelo aditivo para as previsões. No entanto outras estratégias podem ser adotadas como o erro multiplicativo (FIRMINO; NETO; FERREIRA, 2014) ou outras combinações utilizando redes neurais (KHASHEI; BIJARI, 2011). Outra possibilidade é a utilização de meta-learning para o treinamento de um modelo específico para combinar as previsões.

- **Utilização de outros métodos de suavização**: A suavização exponencial simples requer que a série tenha nível constante. No entanto outros métodos podem realizar o mapeamento da tendência como a suavização exponencial dupla (MAKRIDAKIS; WHEELWRIGHT; HYNDMAN, 1998). Esse método pode trazer benefícios para os casos em que a tendência é exponencial ou polinomial.

- **Previsão de séries temporais com *Deep Learning***: algoritmos de *Deep Learning* vêm sido aplicados com sucesso em séries temporais (LäNGKVIST; KARLSSON; LOUTFI, 2014; KUREMOTO et al., 2014). Nesse contexto, é possível avaliar a performance dos algoritmos em resíduos produzidos por outras técnicas.

- **Utilização de *Ensembles***: Ao final do processo de otimização do método proposto, o melhor indivíduo é selecionado para realizar as previsões. No entanto a utilização da combinação de um grupo de previsões (ADEODATO et al., 2011) pode obter melhores resultados. Para isso seria utilizado parte da população final do processo de otimização (ZHOU; WU; TANG, 2002).

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Apêndices

APÊNDICE A – A distributed PSO-ARIMA-SVR hybrid system for time series forecasting

2014 IEEE International Conference on Systems, Man, and Cybernetics
October 5-8, 2014, San Diego, CA, USA

A Distributed PSO-ARIMA-SVR Hybrid System for Time Series Forecasting

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***Abstract*—The combination of techniques in order to achieve more accurate predictions in time series forecasting has been widely applied. Statistical linar models such as the autoregressive integrated moving average (ARIMA) can not capture nonlinear patterns in time series. Therefore nonlinear models such as the support vector regression (SVR) are able to map such patterns. Thus time series can be decomposed in linear and nonlinear patterns. In order to capture both types of patterns a hybrid system comprised by ARIMA and SVR models optimized by the particle swarm optimization (PSO) algorithm is applied to perform predictions. The results show that the proposed method achieved promising results for one-step ahead predictions.**

I. Introduction

Time series forecasting is a challenging task in machine learning and has many applications such as management, stock market indices and production. A time series is a set of observations with a discrete-time stochastic nature. The forecasting of time series is achieved by analyzing past data and then estimating future values.

In order to perform forecasts, a model is used to learn from past data. In the literature there are linear models such as Autoregressive Integrated Moving Average (ARIMA) and Holt-Winters (HW) [1], which are able to map linear patterns from the time series. Artificial Neural networks (ANN) [2]–[5] and support vector machines (SVM) [6], [7] are examples of nonlinear methods.

A time series may present trend or seasonal patterns which can be modeled by traditional statistical methods (ARIMA and HW), however nonlinear patterns are not easily captured. Since real world processes often generate nonlinear time series, the employment of nonlinear models could be more appropriate to increase the accuracy of the system. In this work we focus on SVM models, specifically, the support vector regression (SVR) which is the former approach for regression problems. The SVR is a learning algorithm based on statistical learning theory [8] which employs the structural risk minimization and is capable of performing generalizations based on past data and handling noise.

Both linear and nonlinear methodologies can complement each other on mapping patterns in time series, thus improving the quality of predictions. Previous work of Bates and Granger [9] performed several combinations of architectures for time series prediction. Combinations employing ARIMA and SVR models were investigated by Pai and Lin [10], Xuemei *et al.* [11] and Zhu and Wei [12] in different applications of time series forecasting.

One challenging problem in time series modeling using SVR is to select model parameters correctly, which is very important to improve the accuracy of the forecasts. A traditional approach recommended by Hsu *et al.* [13] is to perform a "grid search" through a range of possible parameters, however this strategy may be time consuming for a wide range of parameters an also may lead to poor generalization capacity of the model. In the literature, systems based on stochastic search were employed to optimize parameter selection. Hsieh *et al.* used a Particle Swarm Optimization (PSO) [14] approach to select SVR parameters in financial time series forecasting. Alwee *et al.* [15] used a hybrid ARIMA-SVR system optimized by two executions of the PSO algorithm, one for the selection of ARIMA parameters and the other for selection of SVR parameters. Huand and Dun [16] proposed a distributed PSO-SVM system with feature selection applied in classification problems.

In this work we propose a distributed ARIMA-SVR hybrid system simultaneously optimized by a discrete PSO [17] and a continuous-valued PSO in order to improve the quality of predictions. The discrete PSO will select the degrees of the Autoregressive (AR), Integrated (I) and Moving Average (MA) which will compose the ARIMA model, while the parameters for the SVR are optimized using the continuous-valued PSO.

The remainder of this paper is organized as follows: Section 2 presents a brief review of the related work, Section 3 shows the proposed method. Section 4 presents the experiments and the conclusions and future works are presented in section 5.

II. Related Work

This section presents works related to the proposed system.

A. Support Vector Regression

The SVR method was proposed by Vapnik [8] and it is based on the structured risk minimization (SRM) principle which performs the minimization of the upper bound of the generalization error. Let $\{\mathbf{x}_i, y_i\}_{i=1}^{l}$ be a training set where $\mathbf{x} \in \Re^d$ is the *i*th input vector, $y_i \in \Re$ is the *i*th prediction output of x_i, d is the embedding dimension of the time series and l is the number of training samples. The objective of SVR is to find the best function from a set of possible functions in

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the form:

$$\{f|f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + \mathbf{b}, w \in \Re^d, b \in \Re\} \quad (1)$$

Where w is the weight vector estimated by minimizing the regularized risk function showed on eq. 2 and $\mathbf{b}$ is a bias or threshold.

$$\frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{l} L(y_i, f(\mathbf{x}_i)) \quad (2)$$

In order to find the best function f, it is necessary to minimize the regularized risk, where $C > 0$ is a regularization factor, $\|.\|$ is a 2-norm and $L(.,.)$ is a loss function. In order to induce sparsity in SVR the ϵ-insensitive loss function presented on eq. 3, which creates an ϵ-tube allowing some predictions to fall within the boundaries of this ϵ-tube.

$$L(y, f(\mathbf{x})) = \begin{cases} 0, & |f(\mathbf{x}) - y| < \epsilon \\ |f(\mathbf{x}) - y| - \epsilon, & \text{otherwise} \end{cases} \quad (3)$$

With the ϵ-insensitive function the SVR is formulated as the minimization of $\frac{1}{2}||\mathbf{w}||^2 + C\sum_{i=1}^{l}(\xi_i - \xi_i{}^*)$, subject to

$$\begin{cases} \mathbf{w}^T\mathbf{x_i} + b - y_i \leq \epsilon + \xi_i \\ y_i - \mathbf{w}^T\mathbf{x_i} - b \leq \epsilon + \xi_i{}^* \\ \xi_i, \xi_i{}^* \geq 0, i = 0...l \end{cases} \quad (4)$$

Where ξ and ξ^* are slack variables used to measure the errors occurred by values outside the boundries determined by the ϵ-tube.

The use of kernels allow the SVR to perform nonlinear mappings into a higher dimensionality space. The regression function takes the form

$$f(\mathbf{x}) = \sum_{i=1}^{l}(\alpha_i - \alpha_i^*)k(\mathbf{x}_i, \mathbf{z}) + \mathbf{b} \quad (5)$$

Where α and α^* are lagrange multipliers and $k(\mathbf{x}_i, \mathbf{x})$ is a kernel function. In this study, the gaussian kernel is adopted ($k(\mathbf{x}_i, \mathbf{x}_j) = exp(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\gamma^2})$), where γ is a parameter of the gaussian kernel.

B. Autoregressive Integrated Moving Average

The ARIMA model was first introduced by Box and Jenkins [18] and has been widely applied in time series forecasting. The predictions in the ARIMA model are linear combinations of past values and past residual errors, and has the form presented on eq. 6.

$$\theta_p(\mathbf{B})(1 - \mathbf{B})^s x_t = \phi_q(\mathbf{B})\epsilon_t \quad (6)$$

where x_t is the actual value and ϵ_t is the residual error at time t, $\mathbf{B}$ is a backward shift operator ($\mathbf{B}x_t = x_{t-1}$ and $\Delta x_t = x_t - x_{t-1} = (1 - \mathbf{B})x_t$), ϕ_i and θ_j are the coefficients, p and q are degrees for the autoregressive and moving average polynomial functions. The integration part is due to the differencing performed to the time series in order to obtain a stationary series. The number of differences is determined by the parameter s.

The ARIMA model is adapted from the structure of the data. However, the presence of nonlinear patterns in the data will limit the capacity of the ARIMA.

C. Particle Swarm Optimization

The PSO algorithm is a population based stochastic search algorithm inspired in social behavior among individuals. Each particle represents a potential solution to the optimization problem and moves in a k dimensional search space. Through each iteration the particles perform movements based on previous knowledge and interactions with other particles.

Each particle maintains a record of its best solution in a vector called $pbest$, and the best solution among all particles is called $gbest$. The particles move in direction of the global best solution and its personal best solution.

To perform movements each particle calculates its velocity and updates its position for each iteration. Consider $pbest_i$, $gbest$ and g_i vectors representing the particles personal best solutions, global best solution and current solution. The velocity v_i is updated as follows:

$$v_i(t+1) = w \times v_i(t) + c_1 \times rnd() \times (pbest_i - g_i(t)) + c2 \times rnd() \times (gbest - g_i(t)) \quad (7)$$

Where $v_i \in [-v_max, v_max]$, rnd() is a random function in the range [0,1], the c_1 and c_2 constants represent the personal and social learning factors and w is the inertia weight. The new solution is the calculated as follows:

$$g_i(t+1) = g_i(t) + v_i(t+1) \quad (8)$$

1) Binary PSO: The discrete PSO was proposed by Kennedy and Eberhart [17] to operate on binary search spaces. Each particle $\mathbf{x_i}$ dimension of a position can take on the binary value 0 or 1 ($\mathbf{g_i} \in B^k$).

The calculation of the velocity showed on eq. 7 remains unchanged, except that g_i, $pbest_i$ and $gbest$ belong to the binary domain and v_i must be constrained to the [0,1] interval as well. The constriction is obtained by using a sigmoid function $Sig(g_i)$, and a new position is calculated as follows:

$$\text{if rnd()} < Sig(v_{i,d}), \text{ then } g_{i,d} = 1; \text{else } g_{i,d} = 0; \quad (9)$$

Where $Sig(v) = \frac{1}{1+e^{-v}}$. The sigmoid functions perform a natural normalization of velocities.

III. Proposed Method

In order to overcome natural limitations of modeling data with single models, this hybrid system incorporates the linear and nonlinear mappings of the time series through a composition of a ARIMA model and SVR optimized by a PSO approach. We can consider a time series as the composition of a linear autocorrelation structure and a nonlinear component [19],

$$x_t = n_t + l_t \quad (10)$$

Where, x_t is a vector with data correlated in time, n_t is the nonlinear component and l_t is the linear component.

The first step of the proposed method is to estimate the linear component through a $ARIMA(p, s, q)$ model producing a residual $e_t = x_t - x_t{}'$ where $x_t{}'$ is the estimation of the linear component. Next, the residuals are embedded with a predefined dimension d where $y_i = [e_i, e_{i+1}, e_{i+2}, ..., e_{i+d}]$

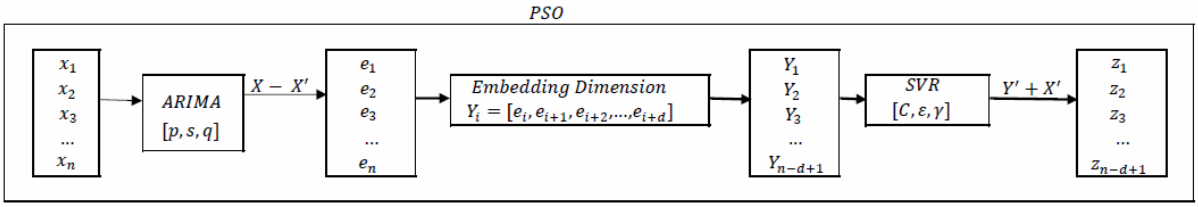


Fig. 1: Execution Flow for the PSO-ARIMA-SVR system

and then modeled by the SVR, which will extract the nonlinear patterns that were not mapped previously. The final forecast will be the combination of forecasts by the two models as follows:

$$z_t = x_t{}' + y_t{}' \tag{11}$$

The execution flow of the algorithm is shown on figure 1

In order to select the best parameter set for both ARIMA and SVR, a discrete PSO is employed along with a continous-valued PSO. The codifications for each particle is showed in fig. 2. For the codification of the discrete part which selects the best $ARIMA(p, s, q)$ model, we use 5 bits. The first 2 bits select the degree of the AR(p) model, the third bit is for the integration step, and the last 2 bits represent the degree of the MA(q) model. The search space is restricted to the binary domain where the only possible are in the $\{0, 1\}$ set.

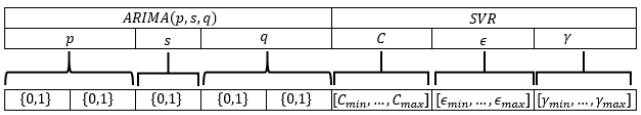


Fig. 2: Encodings for the Particles

The continuous-valued PSO selects the best set of parameters for the SVR model which are the constant C, the regression parameter ϵ and the kernel constant γ. The search space is restricted by minimum and maximum values assigned to the parameters.

Fitness evaluation is based on the final prediction z_t of the system. To assess the quality of prediction we employ the mean square error (MSE) described in eq. 12. A validation set is used to calculate the fitness of a particle, and a test set is used to evaluate the performance of the system.

$$f(g_i) = \sum_{i=1}^{n} \frac{(z_{t,i} - x_t)^2}{n} \tag{12}$$

Algorithm 1 describes the execution of the distributed PSO algorithm.

IV. Experiments

Our proposed approach is compared with other methods such as traditional SVR, ARIMA-SVR and PSO-SVR. For all approaches the SVR implementation was accomplished by using the LIBSVM package [20] in a MATLAB 8.2 environment. The datasets were partitioned, using 50% for training, 25% for validation and 25% for testing.

Algorithm 1 Distributed PSO-ARIMA-SVR

```
count ← 0
Initialize population G, ∀g_i ∈ G
while count < maxIterations do
    while i < N do
        Calculate fitness fit_i = f(g_i)
        pbest_i ← fit_i
        i++
    end while
    Choose the particle with the best fitness as the gbest
    i = 0
    while i < N do
        Calculate the velocity for both discrete and continous PSO using eq. 7
        Update g_i using eq. 8 for the continous parameters and eq. 9 for the discrete
        ones.
        pbest_i ← fit_i
        i++
    end while
end while
Select the best solution in the swarm
```

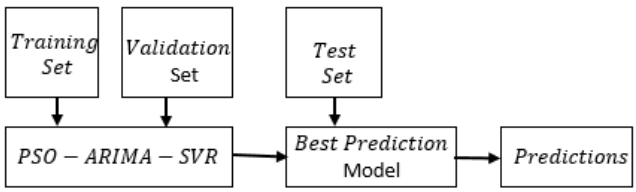


Fig. 3: Experiments Flow

In the traditional SVR method the parameter selection was performed using a grid search over a fixed interval of possible values for the parameters. The values assessed for C, ϵ and γ were in the range $\{10^{-3}, 10^{-2}, 10^{-1}, ..., 10^{2}\}$, $\{10^{-8}, 10^{-7}, ..., 10^{-2}\}$ and $\{2^{-5}, 2^{-4}, ..., 2^{2}\}$ respectively. The best model was selected based on the performance on the test set.

For the ARIMA-SVR approach the p, q and s parameters of the ARIMA(p,s,q) model are defined with grid search in the training data with parameters ranging from $\{0, 1, 2\}$, $\{0, 1\}$ and $\{0, 1, 2\}$ respectively and the best ARIMA(p,s,q) model is selected based on the MSE on the validation set. The SVR parameter selection for the ARIMA-SVR method is the same as employed in the traditional SVR.

In the PSO-SVR method, the PSO algorithm performs a stochastic search to find the best set of parameters of SVR. The PSO algorithm is executed for 50 iterations, the inertia weight $w = 1$, personal and social learning factors $c_1 = 2$ and $c_2 = 2$ respectively. The search bounds for C, ϵ and γ are within the intervals $[10^4, 10^{-2}]$, $[10^{-1}, 10^{-8}]$ and $[10^3, 10^{-2}]$ respectively.

Dataset	Size	Seasonal	Trend
Sunspot	289	Yes	No
Airline	144	Yes	Yes
IBM	369	No	Yes
Common	99	No	Yes
Paper	120	Yes	Yes

TABLE I: Datasets description

Dataset	Lags
Sunspot	20
Airline	30
IBM	20
Common	10
Paper	20

TABLE II: Best embedding dimension for each dataset

The PSO-ARIMA-SVR is similar to the same experimental setup of the PSO-SVR method, except for the ARIMA parameters (p,s,q) which the search is conducted by the discrete PSO with the following values $\{0, 1, 2\}$, $\{0, 1\}$ and $0, 1, 2$ respectively.

The size of the embedding dimension is a key factor to perform good predictions. In order to determine a suitable embedding dimension of the time series for the SVR model, preliminary executions were performed with dimensions $d = \{10, 20, 30\}$ and the results are presented on table II. The best results on the test set are displayed along with the embedding dimension assigned.

The experiments were performed in five benchmark datasets from the Time series data library [21], including Sunspot, Airline Passengers, IBM stock prices, common stock prices and Paper. The 5 datasets used in this work presents different patterns and sizes. Table I summarizes some characteristics from the datasets and figure 4 presents the autocorrelograms for all datasets.

All datasets were scaled to the $[0, 1]$ range using the eq. 13.

$$x' = \frac{x - min_x}{max_x - min_x} \tag{13}$$

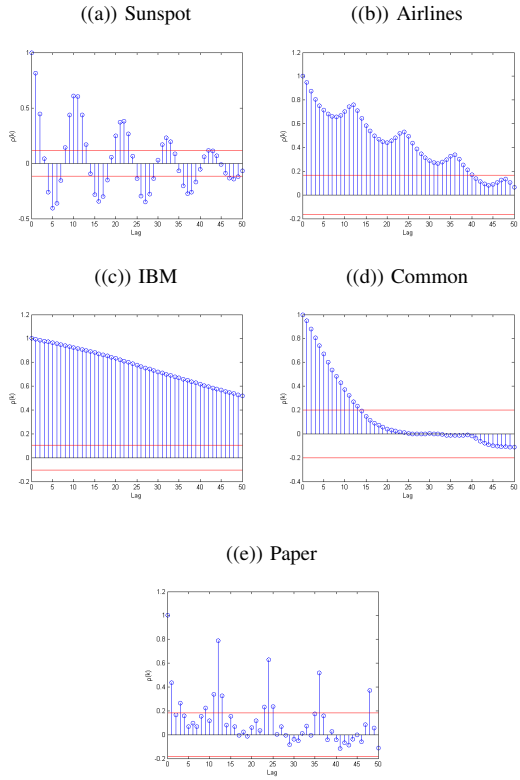
Where min_x is the minimum value of the series x and max_x is the maximum value.

All approaches perform one-step ahead predictions. The experiments were executed for 30 iterations, and the mean MSE and the standard deviation can be analyzed on table III. The results were assessed based on a empirical analysis, a hypothesis testing (A Wilcoxon signed-rank 95% of confidence) and the residual analysis based on the mean value of the residual error obtained from the predictions from all iterations.

A. Results for Sunspot dataset

Based on the results presented on table III the PSO-SVR method achieved the best MSE. However this value is not statistically superior to the PSO-ARIMA-SVR, which outperformed the SVR and ARIMA-SVR. According to the residual analysis on figure 5, we can see that the proposed technique did not capture all patterns in the time series which can be observed by the correlation values out the interval of confidence in very few lags, not characterizing white noise.

Fig. 4: Autocorrelograms for the datasets



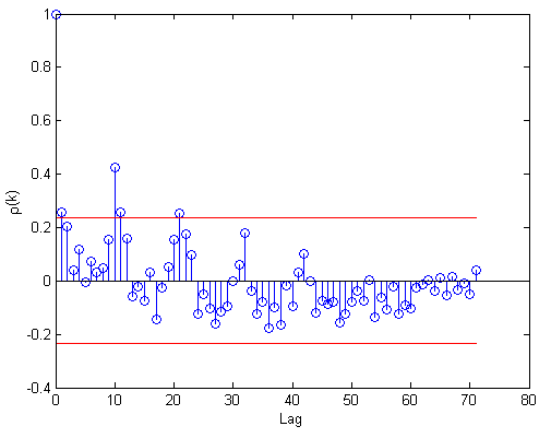


Fig. 5: Autocorrelogram of the Residual Error for Sunspot dataset

B. Results for Airline dataset

In the Airline dataset the PSO-ARIMA-SVR outperformed all methods, achieving statistically superior results. The residual analysis on figure 6 shows that the PSO-ARIMA-SVR successfully modeled the series since the residual has a white noise behavior.

Method	Sunspot	Airline	IBM	common	Paper
SVR	0.0470(0.0000)	0.1908(0.0000)	0.4903(0.0000)	0.2157(0.000)	0.0814(0.0000)
ARIMA-SVR	0.0698(0.0000)	0.0206(0.0000)	0.0935(0.0000)	**0.1049(0.0000)**	0.0102(0.0000)
PSO-SVR	**0.0151(0.0006)**	0.0059(0.0063)	0.0012(0.04×10^{-4})	0.1125(0.0774)	0.0095(0.0020)
PSO-ARIMA-SVR	0.0152(0.0010)	**0.0020(0.0003)**	**0.0008(0.7×10^{-4})**	0.1107(0.0567)	**0.0052(0.0014)**

TABLE III: Mean square errors and Standard deviations for all datasets

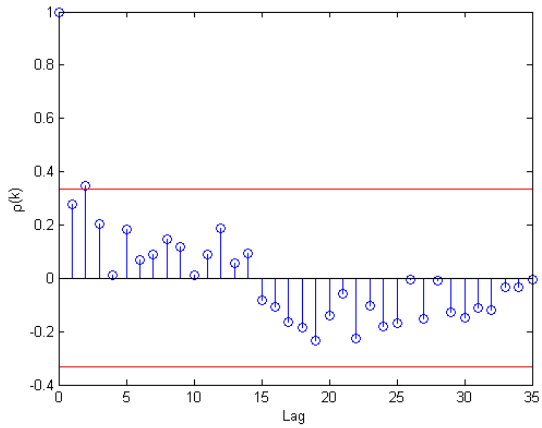


Fig. 6: Autocorrelogram of the Residual Error for the Airline dataset

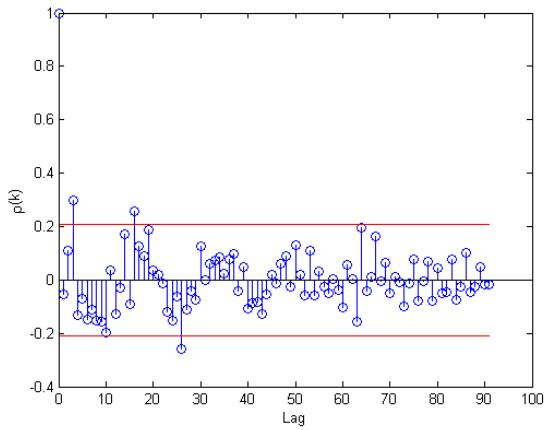


Fig. 7: Autocorrelogram of the Residual Error for the IBM dataset

C. Results for IBM dataset

In the IBM dataset the PSO-ARIMA-SVR approach achieved the best results, and the residual analysis on figure 7 on three lag values the correlation values were significant, thus the residual did not present white noise behavior.

D. Results for common dataset

In the common dataset both ARIMA-SVR, PSO-SVR and PSO-ARIMA-SVR achieved similar performance, outperforming the traditional SVR approach. The residual (figure 8) presented trend patterns, thus the PSO-ARIMA-SVR could not properly model this dataset.

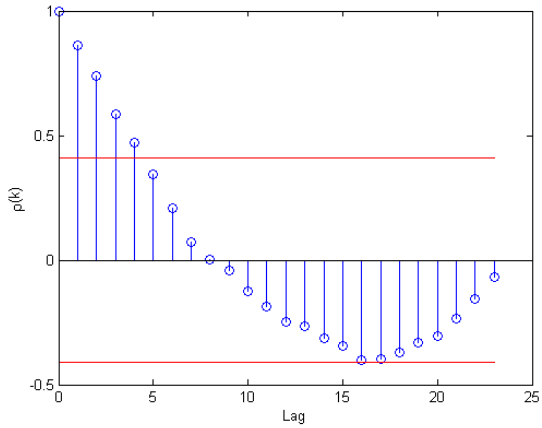


Fig. 8: Autocorrelogram of Residual Error for the Common Dataset

E. Results for Paper dataset

In the paper dataset the PSO-ARIMA-SVR outperformed all techniques, and the residual analysis on figure 9 produced a white noise.

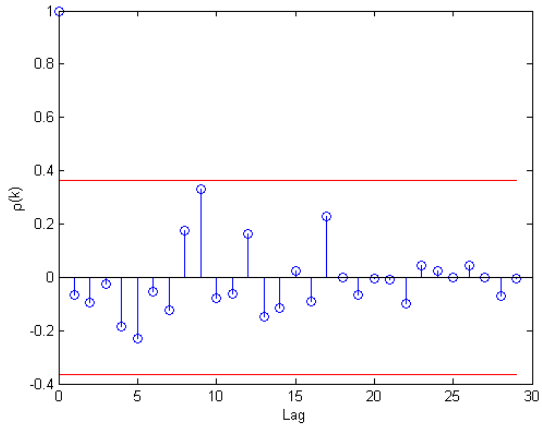


Fig. 9: Autocorrelogram of the Residual Error for the Paper Dataset

V. Conclusion

This work proposed a hybrid system comprised by an Autoregressive Integrated Moving Average (ARIMA) model

and a Support Vector Regression model optimized by the Particle Swarm Optimization (PSO) algorithm to perform one-step ahead predictions. The search process of the PSO is performed through discrete and continuous spaces simultaneously in order to select the best set of parameters. The system captures linear and nonlinear patterns in time series through the use of an ARIMA and SVR model respectively, which complement each other to give more accurate results.

The PSO-ARIMA-SVR method achieved best results in three out of five datasets, and demonstrated promising performance in modeling time series patterns. Specifically, on the datasets that presented trend patterns, the method performed best. Under these settings the hybridization increased the performance of the method and in some datasets achieved similar results in comparison with the other techniques.

The sunspot dataset presents a strong seasonal component demonstrated by positive and negative correlations and the SVR module of the system was not able to map all patterns from this data since is responsible for mapping nonlinear patterns as well, a linear seasonal method (Seasonal ARIMA or Holt-Winters) could be explored in the future by mapping linear trend and seasonality patterns in the time series. The common dataset demonstrates trend patterns in the first lags, however in later lags it does not show significant correlations.

In addition, the embedding dimension was restricted to 10, 20 and 30 lags. For future works, we intend to analyze the performance of a hybridization to select the best lags in a time series along with the optimization of an ARIMA-AVR model. Ensemble approaches will also be investigated in the long-term time series forecasting context.

ACKNOWLEDGMENT

The authors would like to thank FACEPE, CNPq and CAPES (Brazilian Research Agencies) for their financial support.

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APÊNDICE B – A Hybrid Evolutionary System for Parameter Optimization and Lag Selection in Time Series Forecasting

A Hybrid Evolutionary System for Parameter Optimization and Lag selection in Time Series Forecasting

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***Abstract*—The accuracy of time series forecasting can be increased by the employment of evolutionary systems. The improvement in the precision of such systems impact positively on the decision making process of many organizations. In this work we explore the decomposition of time series into linear and nonlinear patterns by the use of an autoregressive integrated moving average (ARIMA) method and a support vector machine (SVM) method using a particle swarm optimization algorithm to simultaneously perform parameter selection and lag selection. Experiments were performed using datasets from the time series data library amd the results demonstrated that the proposed method achieved promising results for one-step ahead predictions.**

I. Introduction

Time series forecasting is an important task in the field of machine learning and has applications in diverse areas including stock market, hydrology and business. A time series is a sequence of data points correlated in time driven by a dynamical system. Linear time series can be modeled by traditional statistical methods, some of which are: Autoregressive integrated moving average (ARIMA) and Holt-Winters [1]. However many of the real world time series present nonlinear patterns which cannot be mapped by the aforementioned methods.

Nonlinear patterns in time series are best modeled by nonlinear methods such as artificial neural networks (ANN) [2] and support vector regression (SVR) [3]. The performance of these techniques depend on the quality of their parameters for a given problem. One possible strategy for parameter setting is to perform trial and error approaches, which are very time consuming. Another, is to employ optimization techniques including particle swarm optimization (PSO) [4] and genetic algorithms (GA) [5] to search for the best set of parameters. Donate *et al.* [2] used an evolutionary ANN to build ensembles. Cortez *et al.* employed a GA to optimize ANN parameters in time series forecasting [6], Wen *et al.* [7]used a Immune evolutionary algorithm and ANN for time series prediction.

According to Zhang [8] time series can be considered as the composition of both linear and nonlinear patterns, thus the combination of linear and nonlinear methods can improve the accuracy of the forecasting system [9]–[11].

An important parameter in time series forecasting is the embedding dimension of lags used in the construction of the model. The embedding theorem of Takens [12] showed that for a time series with data generated by a dynamical system it is possible to model the series using a time-delay embedding with a sufficient large dimension. Therefore the dimension size m and the lag τ have to be determined previously with the purpose of improving the quality of the model. Some methods in the literature were proposed for the optimal selection of embedding parameters [13]–[19] however there is not a single best method for all situations. Among the methods for detecting embedding dimensions, two will be given emphasis: the uniform embedding and the nonuniform embedding [20]. The uniform embedding is based on the assumption that the lag τ is the same in all dimensions, although this method is effective for classic chaotic systems it performs poorly in time series with multiple time scales. The nonuniform embedding allows different lag values in all dimensions and is able to improve performance over the former approach in time series with multiple scales.

In order to perform linear and nonlinear mappings of time series patterns with an automatic parameter and lag selection, the contribution of this work is to employ a hybrid system comprised by an evolutionary ARIMA-SVR system with lag selection to perform forecasts. In this system the parameters of the ARIMA(p, s, q), SVR(C, ϵ, γ) and the lags of the time series are simultaneously selected by a PSO algorithm which uses both discrete [21] and continuous versions of PSO. The lag selection is performed initially as the uniform embedding, however the lags are selected by the PSO according with the performance of the system which is measured based on one-step-ahead predictions.

The remainder of the paper is organized as follows: Section II presents an overview of the related work, section III contains the proposed method architecture, section IV explains the experimental setup and discussion of results, the conclusion is presented on section V.

II. Related Work

In this section we present a brief review of lag selection methods, SVR, ARIMA and PSO.

A. Lag Selection

Time series forecasting is based on past values in order to perform predictions, thus the behavior of past data must be

978-1-4799-5618-0/14 $31.00 © 2014 IEEE
DOI 10.1109/BRACIS.2014.24

taken into consideration. Let $X_t = x_t \in \Re | t = 1, 2, 3...N$ be a scalar time series, the embedding process converts a scalar time series into a vector $u_t \in \Re^d$ for some integer d, where d is the embedding dimension.

The uniform embedding dimension approach takes into consideration a vector $u_t = (x(t), x(t-\tau), x(t-2\tau), x(t-3\tau), ..., x(t-(d-1)\tau))$ with fixed value of τ for all dimensions in the interval $[1...d]$. Nonuniform embedding consists of a vector $u_t = (x(t), x(t-\tau_2), x(t-\tau_3), x(t-\tau_4), ..., x(t-\tau_d))$ where each dimension has a value for the lag τ_d. The latter approach is effective in real world time series with many periodicities and noise.

A lag selection approach proposed by Ribeiro *et al.* [22] used a Frankenstein's PSO [23] to select the best lags in a time series. Considering a time series $T = \{4, 2, 5, 3, 6, 4, 7\}$, and an embedding dimension $d = 5$, and one-step-ahead predictions. The PSO will encode the solution as a vector $V = \{0, 1, 0, 0, 1\}$ where each value will represent the presence or absence of a given lag in a prediction process. For instance, the prediction of the sixth value 4, we will take into consideration the vector of past data $\{2, 6\}$. The process performs uniform embedding with lag $\tau = 1$, but the lags are then chosen in a nonuniform fashion by the PSO algorithm. The advantage of this approach is the dimensionality reduction which eliminates unnecessary lag values and increases the computational efficiency by reducing the training set [24].

B. Support Vector Regression

The SVR method was proposed by Vapnik [25] and it is based on the structured risk minimization (SRM) principle which performs the minimization of the upper bound of the generalization error. Let $\{\mathbf{x}_i, y_i\}_{i=1}^{l}$ be a training set where $\mathbf{x} \in \Re^d$ is the *i*th input vector, $y_i \in \Re$ is the *i*th prediction output of x_i, d is the embedding dimension of the time series and l is the number of training samples. The objective of SVR is to find the best function from a set of possible functions in the form:

$$\{f | f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b, w \in \Re^d, b \in \Re\} \tag{1}$$

Where w is the weight vector estimated by minimizing the regularized risk function showed on eq. 2 and $\mathbf{b}$ is a bias or threshold.

$$\frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{l} L(y_i, f(\mathbf{x}_i)) \tag{2}$$

In order to find the best function f, it is necessary to minimize the regularized risk, where $C > 0$ is a regularization factor, $\|.\|$ is a 2-norm and $L(.,.)$ is a loss function. In order to induce sparsity in SVR the ϵ-insensitive loss function presented on eq. 3, which creates an ϵ-tube allowing some predictions to fall within the boundaries of this ϵ-tube.

$$L(y, f(\mathbf{x})) = \begin{cases} 0, & |f(\mathbf{x}) - y| < \epsilon \\ |f(\mathbf{x}) - y| - \epsilon, & \text{otherwise} \end{cases} \tag{3}$$

With the ϵ-insensitive function the SVR is formulated as the minimization of $\frac{1}{2}||\mathbf{w}||^2 + C\sum_{i=1}^{l}(\xi_i - \xi_i{}^*)$, subject to

$$\begin{cases} \mathbf{w}^T\mathbf{x_i} + b - y_i \leq \epsilon + \xi_i \\ y_i - \mathbf{w}^T\mathbf{x_i} - b \leq \epsilon + \xi_i{}^* \\ \xi_i, \xi_i{}^* \geq 0, i = 0...l \end{cases} \tag{4}$$

where ξ and ξ^* are slack variables used to measure the errors occurred by values outside the boundaries determined by the ϵ-tube.

The use of kernels allow the SVR to perform nonlinear mappings into a higher dimensionality space. The regression function takes the form

$$f(\mathbf{x}) = \sum_{i=1}^{l} (\alpha_i - \alpha_i^*) k(\mathbf{x}_i, \mathbf{z}) + b \tag{5}$$

where α and α^* are lagrange multipliers and $k(\mathbf{x}_i, \mathbf{x})$ is a kernel function. In this study, the gaussian kernel is adopted ($k(\mathbf{x}_i, \mathbf{x}_j) = exp(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\gamma^2})$), where γ is a parameter of the gaussian kernel.

C. Autoregressive Integrated Moving Average

The ARIMA model was first introduced by Box and Jenkins [26] and has been widely applied in time series forecasting. The predictions in the ARIMA model are linear combinations of past values and past residual errors, and has the form presented on eq. 6.

$$\theta_p(\mathbf{B})(1 - \mathbf{B})^s \mathbf{x}_t = \phi_q(\mathbf{B})\epsilon_t \tag{6}$$

where x_t is the actual value and ϵ_t is the residual error at time t, $\mathbf{B}$ is a backward shift operator ($\mathbf{B}x_t = x_{t-1}$ and $\Delta x_t = x_t - x_{t-1} = (1 - \mathbf{B})x_t$), ϕ_i and θ_j are the coefficients, p and q are degrees for the autoregressive and moving average polynomial functions. The integration part is due to the differencing performed to the time series in order to obtain a stationary series. The number of differences is determined by the parameter s.

The ARIMA model is adapted from the structure of the data. However, the presence of nonlinear patterns in the data will limit the capacity of the ARIMA.

D. Particle Swarm Optimization

The PSO algorithm is a population based stochastic search algorithm inspired in social behavior among individuals. Each particle represents a potential solution to the optimization problem and moves in a k dimensional search space. Through each iteration the particles perform movements based on previous knowledge and interactions with other particles.

Each particle maintains a record of its best solution in a vector called $pbest$, and the best solution among all particles is called $gbest$. The particles move in direction of the global best solution and its personal best solution.

To perform movements each particle calculates its velocity and updates its position for each iteration. Consider $pbest_i$, $gbest$ and g_i vectors representing the particles personal best solutions, global best solution and current solution. The velocity v_i is updated as follows:

$$\begin{aligned} \mathbf{v}_i(t+1) = w \times \mathbf{v}_i(t) + c_1 \times rnd() \times (\mathbf{pbest}_i - \mathbf{g}_i(t)) \\ + c2 \times rnd() \times (\mathbf{gbest} - \mathbf{g}_i(t)) \end{aligned} \tag{7}$$

where $\mathbf{v}_i \in [-v_{max}, v_{max}]$, rnd() is a random function in the range [0,1], the c_1 and c_2 constants represent the personal and

social learning factors and w is the inertia weight. The new solution is calculated as follows:

$$\mathbf{g}_i(t+1) = \mathbf{g}_i(t) + \mathbf{v}_i(t+1) \tag{8}$$

The discrete PSO was proposed by Kennedy and Eberhart [21] to operate on binary search spaces. Each particle $\mathbf{x_i}$ dimension of a position can take on the binary value 0 or 1 ($\mathbf{g_i} \in B^k$).

The calculation of the velocity showed on eq. 7 remains unchanged, except that $\mathbf{g}_i$, $\mathbf{pbest}_i$ and $\mathbf{gbest}$ belong to the binary domain and $\mathbf{v}_i$ must be constrained to the [0,1] interval as well. The constriction is obtained by using a sigmoid function $Sig(\mathbf{g}_i)$, and a new position is calculated as follows:

$$\text{if rnd()} < Sig(\mathbf{v}_{i,d}), \text{ then } \mathbf{g}_{i,d} = 1; \text{else } \mathbf{g}_{i,d} = 0; \tag{9}$$

where $Sig(\mathbf{v_i}) = \frac{1}{1+e^{-\mathbf{v_i}}}$.

III. Proposed Method

A time series can be considered as the composition of a linear autocorrelation structure and a nonlinear component [8],

$$\mathbf{x_t} = \mathbf{n_t} + \mathbf{l_t} \tag{10}$$

where, $\mathbf{x_t}$ is a vector with data correlated in time, $\mathbf{n_t}$ is the nonlinear component and $\mathbf{l_t}$ is the linear component.

This work proposes a hybrid system comprised by linear and nonlinear models, ARIMA and SVR, which performs linear and nonlinear mappings in the time series. Parameter selection and lag selection are performed by a PSO algorithm that combines both continuous and discrete approaches.

The first step of the proposed method is to estimate the linear component through a $ARIMA(p, s, q)$ model producing a residual $\mathbf{e_t} = \mathbf{x_t} - \mathbf{x_t}'$ where $\mathbf{x_t}'$ is the estimation of the linear component. Next, the residual errors are embedded with a predefined dimension d resulting in a vector $\mathbf{y}_i = [e_i, e_{i+1}, e_{i+2}, ..., e_{i+d}]$. The next step is to select the most relevant lags inside the search space of size d, this approach is performed with the discrete PSO. The vector with the selected lags will be the input for the SVR, which will extract the nonlinear patterns that were not mapped previously. The final forecast will be the combination of forecasts by the two models as follows:

$$\mathbf{z_t} = \mathbf{x_t}' + \mathbf{y_t}' \tag{11}$$

The execution flow of the algorithm is shown on figure 1.

In order to select the best set of lags and best parameter set for both ARIMA and SVR, a discrete PSO is employed along with a continous-valued PSO. The codifications for each particle is showed in Fig. 2. For the codification of the discrete part which selects the best $ARIMA(p, s, q)$ model, we use 5 bits. The first 2 bits select the degree of the AR(p) model, the third bit is for the integration step, and the last 2 bits represent the degree of the MA(q) model. The decoding process for the $ARIMA$ is performed by a summation of bits (e.g. if the five bits were $\{01011\}$, the decoding would result in a $ARIMA(1, 0, 2)$ model). A feature mask is used represent the presence or the absence of a given lag and the dimension of the feature mask will be the same as the predefined embedding dimension of the time series. The search space is restricted to the binary domain where the only possible are in the $\{0, 1\}$ set.

The continuous-valued PSO selects the best set of parameters for the SVR model which are the constant C, the regression parameter ϵ and the kernel constant γ. The search space is restricted by minimum and maximum values assigned to the parameters.

The PSO algorithm performs search in a solution space of dimension $d + 8$, corresponding to the embedding dimension d, 5 bits for the ARIMA parameters and 3 SVR parameters.

Fitness evaluation is based on the final prediction z_t of the system. To assess the quality of prediction we employ the mean square error (MSE) described in eq. 12. A validation set is used to calculate the fitness of a particle, and a test set is used to evaluate the performance of the system.

$$f(g_i) = \sum_{i=1}^{n} \frac{(\mathbf{z}_{t,i} - \mathbf{x}_t)^2}{n} \tag{12}$$

The error analyzed in fitness function is originated from the final predictions of the system. The PSO algorithm runs for a predefined number of iterations and the particle (solution) with best fitness is chosen to perform predictions.

IV. Experiments

Our proposed approach is compared with other methods such as traditional SVR, ARIMA-SVR, PSO-SVR, PSO-SVR with lag selection (PSO-SVR-LS) and a PSO-ARIMA-SVR. The hybrid system of this paper will be abbreviated as PSO-ARIMA-SVR-LS. For all approaches, the SVR implementation was accomplished by using the LIBSVM package [27] in a MATLAB 8.2 environment. The datasets were partitioned, using 50% for training, 25% for validation and 25% for testing.

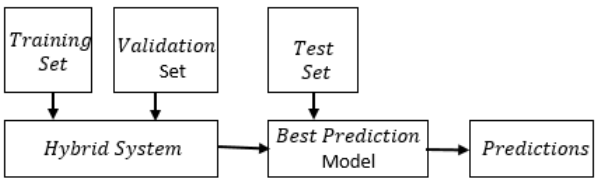


Fig. 3: Experiments Flow

In the traditional SVR method the parameter selection was performed using a grid search over a fixed interval of possible values for the parameters. The values assessed for C, ϵ and γ were in the range $\{10^{-3}, 10^{-2}, 10^{-1}, ..., 10^{2}\}$, $\{10^{-8}, 10^{-7}, ..., 10^{-2}\}$ and $\{2^{-5}, 2^{-4}, ..., 2^{2}\}$ respectively. The best model was selected based on the performance on the test set.

For the ARIMA-SVR approach the p, q and s parameters of the ARIMA(p,s,q) model are defined with grid search in the training data with parameters ranging from $\{0, 1, 2\}$, $\{0, 1\}$ and $\{0, 1, 2\}$ respectively and the best ARIMA(p,s,q) model is selected based on the MSE on the validation set. The SVR parameter selection for the ARIMA-SVR method is the same as employed in the traditional SVR.

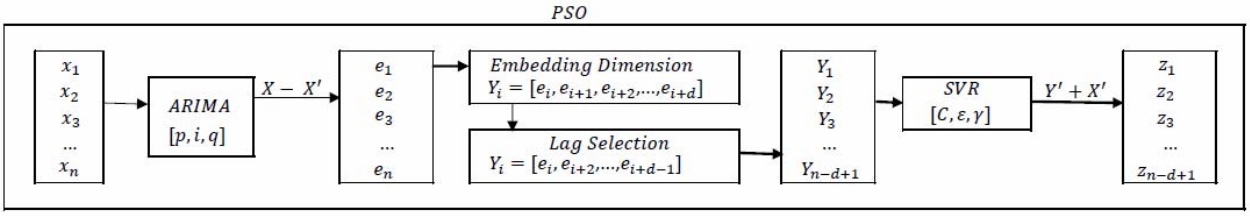


Fig. 1: Architecture of the Hybrid System

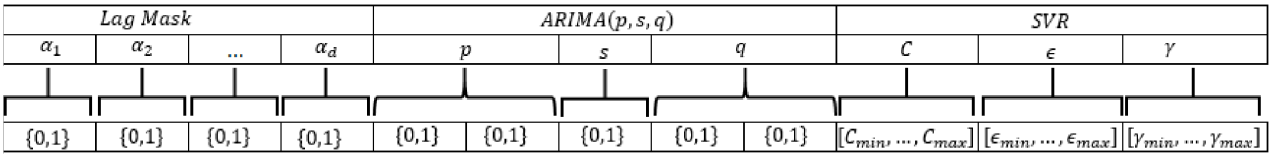


Fig. 2: Codings for the particles in PSO

In the PSO-SVR method, the PSO algorithm performs a stochastic search to find the best set of parameters of SVR. The PSO algorithm is executed for 50 iterations, a population size was set to 30, the inertia weight $w = 1$, personal and social learning factors $c_1 = 2$ and $c_2 = 2$ respectively. The search bounds for C, ϵ and γ are within the intervals $[10^4, 10^{-2}]$, $[10^{-1}, 10^{-8}]$ and $[10^3, 10^{-2}]$ respectively.

PSO-SVR-LS uses same PSO parameters and search bounds for SVR parameters. The lag selection is performed with a predefined dimension d using a discrete PSO where the values for each dimension will be in the $\{0, 1\}$ interval. The PSO-ARIMA-SVR is similar to the same experimental setup of the PSO-SVR method, except for the ARIMA parameters (p, s, q) which the search is conducted by the discrete PSO with the following values $\{0, 1, 2\}$, $\{0, 1\}$ and $0, 1, 2$ respectively. The PSO-ARIMA-SVR-LS method uses the configurations from the PSO-ARIMA-SVR with the lag selection.

The size of the embedding dimension is a key factor to perform good predictions. In order to determine a suitable embedding dimension of the time series for the SVR model, preliminary executions were performed with dimensions $d = \{10, 20, 30\}$ and the results are presented on table II. The best results on the test set are displayed along with the embedding dimension assigned.

The experiments were performed in four benchmark datasets from the Time series data library [28], including Airline Passengers, IBM stock prices, common stock prices, paper sales, Slaughtered pigs and gasoline demand. The 6 datasets used in this work presents different patterns and sizes. Table I summarizes some characteristics from the datasets and figure 4 presents the autocorrelograms for all datasets.

All datasets were scaled to the $[0, 1]$ range using the eq. 13.

$$x' = \frac{x - min_x}{max_x - min_x} \tag{13}$$

Dataset	Description
Passenger	International airline passengers. Jan 1949 to Dec 1960
IBM	IBM stock: daily, 17 May 1961 to 2 November 1962
Common	Annual common stock price, 1871 to 1970
Gas	Monthly gasoline demand 1960 to 1975
Paper	Sales for printing and writing paper Jan 1963 to Dec 1972
Pigs	Monthly number of pigs slaughtered. Jan 1980 to Aug 1995

TABLE I: Datasets description

Dataset	Lags
Passenger	30
IBM	20
Common	10
Gas	10
Paper	20
Pigs	20

TABLE II: Embedding dimension for each dataset

where min_x is the minimum value of the series x and max_x is the maximum value.

All approaches perform one-step ahead predictions. The experiments were executed for 30 iterations, the mean MSE and mean absolute percentage error (MAPE) [22] and the standard deviation can be analyzed on Tables III and IV .

A. Results analysis

The results were assessed based on a empirical analysis, a hypothesis testing (A Wilcoxon signed-rank 95% of confidence) and the residual analysis based on the mean value of the residual error obtained from the predictions from all iterations. If the correlations of the explored lags do not extrapolate the confidence interval then the residual will present a white noise behavior.

Figure 5 shows the autocorrelogram of the mean residual error on the test set produced by the PSO-ARIMA-SVR-LS method. The method could perform satisfactory mappings on

Method	Passenger	IBM	Common	Gas	Paper	Pigs
SVR	0.1908(0.0000)	0.4903(0.0000)	0.2157(0.0000)	0.2154(0.0000)	0.0814(0.000)	0.0151(0.0000)
ARIMA-SVR	0.0206(0.0000)	0.0935(0.0000)	0.1049(0.0000)	0.0312(0.0000)	0.0102(0.0000)	0.0143(0.0000)
PSO-SVR	0.0059(0.0063)	0.0012(4.8E-6)	0.1125(0.0774)	0.0120(0.0066)	0.0095(0.0020)	0.0138(0.0008)
PSO-SVR-LS	0.0098(3.4E-4)	0.0012(7.1E-5)	0.0968(0.0567)	0.0132(0.0010)	0.0092(0.0014)	0.0139(0.0010)
PSO-ARIMA-SVR	0.0020(0.0003)	0.0008(3.6E-6)	0.1107(0.0456)	0.0067(0.0089)	**0.0052(0.0020)**	0.0139 (0.0008)
PSO-ARIMA-SVR-LS	**0.0013(4.4E-4)**	**0.0007(9.4E-5)**	**0.0562(0.0514)**	**0.0055(0.0010)**	0.0075(0.0035)	**0.0124(0.0011)**

TABLE III: Mean square errors and Standard deviations (in parenthesis) for all datasets

Method	Passenger	IBM	Common	Gas	Paper	Pigs
SVR	0.6349(0.0000)	3.8930(0.0000)	0.5944(0.0000)	0.5730(0.000)	0.5775(0.0000)	0.1384(0.0000)
ARIMA-SVR	0.1412(0.0000)	1.5373(0.0000)	0.6215(0.0000)	0.2177(0.0000)	0.2471(0.0000)	0.1398(0.0000)
PSO-SVR	0.1052(0.0431)	0.1507(2.6E-4)	0.3750(0.1557)	0.1118(0.0272)	**0.1437(0.0157)**	0.1300(0.0055)
PSO-SVR-LS	0.1315(0.0675)	0.1508(1.9E-4)	0.3364(0.0669)	0.1151(0.0353)	0.1469(0.0216)	0.1303(0.0074)
PSO-ARIMA-SVR	0.0566(0.0050)	0.1453(0.0023)	0.3630(0.0951)	0.0819(0.0055)	0.1496(0.0252)	0.1307(0.0054)
PSO-ARIMA-SVR-LS	**0.0466(0.0089)**	**0.1304(0.0077)**	**0.2646(0.1122)**	**0.0780(0.0065)**	0.1910(0.0578)	**0.1241(0.0046)**

TABLE IV: Mean absolute percentage error and Standard deviations (in parenthesis) for all datasets.

Fig. 4: Autocorrelograms for the datasets

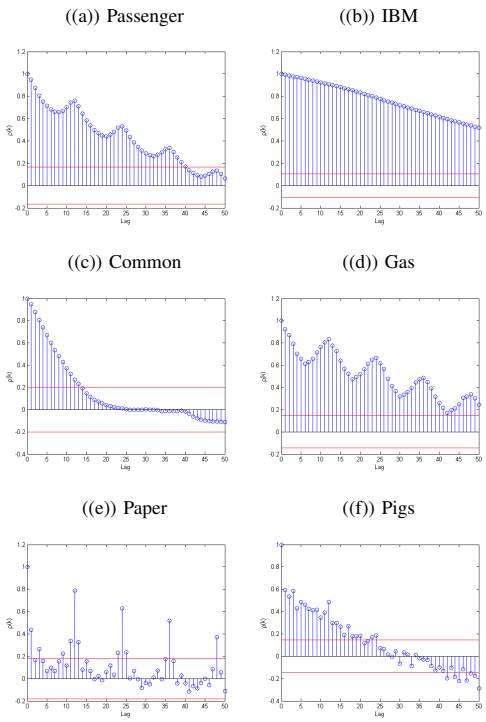
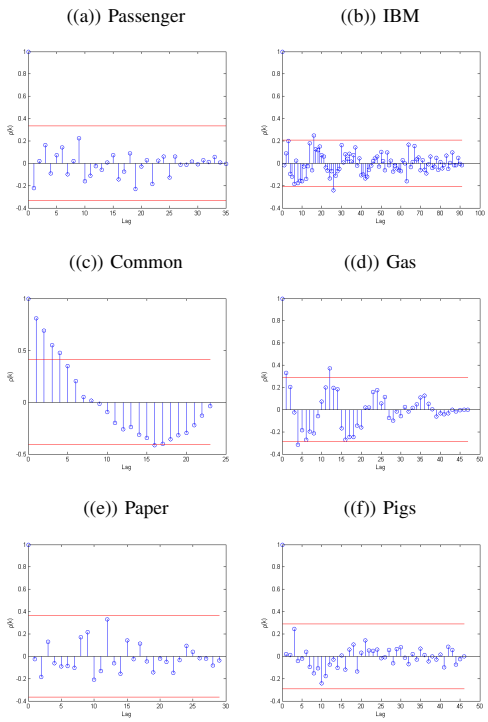


Fig. 5: Residual analysis on the test set for all datasets



most of the time series, however in the Common dataset it showed the presence of trend patterns in the residual error.

Tables III and IV shows the performance of the techniques on the test set for all datasets, and the lowest error values are highlighted.

The Passenger and Gas datasets present trend and seasonal patterns. The PSO-ARIMA-SVR-LS outperformed the other methods for both datasets in terms of MSE and MAPE. The residual error analysis also shows that the proposed technique could map all patterns in the datasets.

In the IBM dataset trend patterns are presented. The proposed technique achieved the lowest error values, however in the residual analysis there are two lags that pointed high correlation, thus indicating that the proposed method could not map all patterns in the time series.

The trend patterns in the Common dataset could not be mapped by the proposed method, however the proposed method achieved the lowest error values (both MSE and MAPE).

In the Paper dataset the best MSE and MAPE values

were achieved by the PSO-ARIMA-SVR and PSO-SVR, respectively. Although the proposed technique was able to map patterns from the time series, it did not outperform the other techniques.

The Pig dataset presents trend patterns and the PSO-ARIMA-SVR-LS achieved the lowest error values and produced white noise residuals.

V. Conclusion

In this work a hybrid evolutionary system for parameter optimization and lag selection in time series forecasting has been proposed. The system is comprised by an autoregressive integrated moving average (ARIMA) model and a support vector regression model and a particle swarm optimization algorithm to find the best set of parameters for both methods and to perform lag selection.

The ARIMA and SVR models perform linear and nonlinear mappings (respectively) in the time series, thus increasing the forecasting capabilities of the system. Another important aspect is the selection of the most relevant lags among a given embedding dimension, this process increases the forecasting capabilities of the system. In order to perform parameter and lag selection, the PSO algorithm is employed using both discrete and continuous valued versions in order to operate in the binary and continuous domains.

The proposed method (PSO-ARIMA-SVR-LS) achieved the best results in 5 out of 6 datasets in regarding the MSE and MAPE performance measures. Residual analysis showed that in the common dataset there were trend patterns discriminated by the autocorrelogram.

Under these settings the hybrid system increased the accuracy of predictions and demonstrated promising results for future work.

In order to further increase performance of the system, ensemble approaches will be employed in the PSO particles. Instead of using only the best particle, many could be used and different combinations strategies could be investigated.

Acknowledgment

The authors would like to thank FACEPE, CNPq and CAPES (Brazilian Research Agencies) for their financial support.

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APÊNDICE C – A hybrid evolutionary decomposition system for time series forecasting

Neurocomputing 180 (2016) 27–34

Contents lists available at ScienceDirect

Neurocomputing

journal homepage: www.elsevier.com/locate/neucom

A hybrid evolutionary decomposition system for time series forecasting

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ARTICLE INFO

Article history:
Received 28 March 2015
Received in revised form
10 July 2015
Accepted 10 July 2015
Available online 10 November 2015

Keywords:
Time series forecasting
Hybrid intelligent systems

ABSTRACT

In the field of time series forecasting, the combination of linear and nonlinear models has been explored to improve the accuracy of predictions when compared with the performance of individual models. Traditional forecasting methods such as the autoregressive integrated moving average (ARIMA) and exponential smoothing (ETS) are employed to map linear patterns in a time series while neural networks (ANNs) and support vector machines (SVMs) map nonlinear patterns. A traditional hybrid ARIMA–ANN system works by performing linear relationships in the data, considering the residual error a nonlinear component which is mapped by the ANN. The nature of a time series can be taken into consideration before applying a model. This work employs both linear and nonlinear patterns of a time series based on its volatility. This is explored using a hybrid evolutionary system composed by a simple exponential smoothing filter, ARIMA and autoregressive (AR) linear models and a SVR model. Particle swarm optimization is employed to optimize the order of the AR model, SVR parameters and number of lags of the time series. Experimental results show that the evolutionary hybrid system presented promising results in the forecasting domain.

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1. Introduction and related work

Time series forecasting has become an important task in the field of machine learning and has applications in diverse areas including stock market, hydrology and business. Through the study of historical observations of a given variable it is possible to model its underlying process. A time series is a sequence of data points correlated in time and driven by a dynamical system. Univariate time series can be represented by a vector $\mathbf{y} = [y_1, y_2, \ldots, y_n]^T$, where y_t is the value measured at time-step t and n is the total number of observations.

Traditional linear models such as the autoregressive integrated moving average (ARIMA) [1] assume that a time series is composed by linear relationship of past data values and errors. In the same way, future values are also considered to be a realization of current and past observations and error terms. An ARIMA model has an autoregressive (AR) component and a moving average (MA) component which take into consideration past observations and past error terms respectively. Box and Jenkins [1] proposed a procedure to determine a suitable number of past observations and errors, often called the order of an ARIMA model. Since real-world time series are often nonlinear [2], it is not advised to assume that the time series is generated by a linear process. ARIMA model can model linear patterns in a time series, however its application in the presence of nonlinear patterns may not be adequate due to this limitation.

Artificial neural networks (ANNs) [3] and support vector regression (SVR) [4] are very important nonlinear techniques in the time series forecasting field. They can perform nonlinear function approximations and also generalize to future data. In order to obtain accurate models for a given problem, parameter setting plays a crucial role. A simple approach for parameter setting is based on trial and error, however underfitting may occur if the parameter space is not large enough or it can be a repetitive time-consuming task. Another approach is to employ optimization techniques including particle swarm optimization (PSO) [5] and genetic algorithms (GA) [6] to search for the best set of parameters. Furthermore, the application of nonlinear techniques in linear time series may produce mixed results [7], thus it is not advised to apply nonlinear techniques in any kind of data without previous analysis.

To overcome natural limitations from the aforementioned class of models and to improve the forecasting accuracy for time series, the combination of linear and nonlinear models in the same problem may be employed, since it is difficult to completely know *a priori* the characteristics of the real-world data in terms of linearity of patterns. In addition, a single model may not be adequate for

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http://dx.doi.org/10.1016/j.neucom.2015.07.113
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every problem, since capturing different patterns equally well is a complex task. In order to address this problem, Zhang [8] proposed a hybrid ARIMA–ANN model to map different patterns in a time series by first capturing linear patterns using the ARIMA model and then employing an ANN on the residual errors. In this sense, the time series is considered a linear composition of linear and nonlinear patterns. Combinations employing ARIMA and SVR models were investigated by Pai and Lin [9], Xuemei et al. [10] and Zhu and Wei [11] in different applications of time series forecasting. Khashei and Bijari [7] explored the decomposition of the time series as a function composed by linear and nonlinear patterns. First an ARIMA model was employed and the residuals along with the original series were used in the ANN model. Babu and Reddy [12] used the nature of a time series to perform a decomposition of a time series in low-volatility and high-volatility components.

Even though these hybrid models are able to outperform each individual component, some studies report inconsistent results [7] due to assumptions made in the model specification stage, resulting in different approximations to the data generating process of a time series. For a given ANN architecture, there are many initial configurations of weights, therefore after the learning process is completed the final configuration may yield a high-bias network [13]. To increase the accuracy of forecasts, hybrid evolutionary systems have been employed. Hsieh et al. [14] used a particle swarm optimization (PSO) [15] approach to select SVR parameters in financial time series forecasting. Alwee et al. [16] used a hybrid ARIMA-SVR system optimized by two executions of the PSO algorithm, one for the selection of ARIMA parameters and the other for selection of SVR parameters. Huand and Dun [17] proposed a distributed PSO-SVM system with feature selection applied in classification problems.

Also an important parameter in time series forecasting is the number of past observations (dimension of lags) used in the construction of the model. The theorem of Takens [18] showed that for a given time series it is possible to model the series using a sufficient large dimension of lags. The dimension size m and the lag τ have to be determined previously with the purpose of improving the quality of the model. Some methods in the literature employ automatic lag selection [19–23] however there is not a single best method for all situations. From an ANN point of view, the input lag structure along with the number of hidden nodes and connection weights are essential for forecasting accuracy [13]. The SVR is very sensitive to its hyper-parameters setting, thus a set of hyper-parameters must be found and the complexity of the training model can be reduced with a suitable lag structure [24].

In this paper a new hybrid evolutionary ARIMA-SVR model is proposed based not only on the decomposition of the time series in linear and nonlinear patterns but also taking into consideration the nature of a time series. A simple exponential smoothing (ETS) filter is employed to decompose the time series into low-volatility and high-volatility components, then an ARIMA model is applied to the low-volatility component, and the high-volatility component is decomposed into linear and nonlinear patterns by an AR-SVR hybrid model. The parameter selection and lag selection is performed by a PSO algorithm. The final forecast will be the summation of the predictions from each component, resulting in a more general and accurate model. Experiments are performed on 12 datasets from the Time Series Data Library [25] and one electricity price data from the Australian National Electricity Market [12]. The results show that the proposed method outperform the models of Babu and Reddy [12], Oliveira and Ludermir [23] and individual models (ARIMA, and Exponential Smoothing).

The remainder of this paper is organized as follows: Section 2 presents a brief description of some techniques employed in this work. In Section 3 the proposed method architecture is described. The experimental setup is presented in Section 4 and the conclusions and final remarks are presented in Section 5.

2. Literature review

Some methods used in this work are briefly described in this section.

2.1. Support vector regression

The SVR method was proposed by Vapnik [26] and it is based on the structured risk minimization (SRM) principle which performs the minimization of the upper bound of the generalization error through the use of a subset of data points called support vectors. Let $\{\mathbf{x}_i, y_i\}_{i=1}^{l}$ be a training set where $\mathbf{x} \in \Re^d$ is the ith input vector, $y_i \in \Re$ is the ith prediction output of x_i, d is the embedding dimension of the time series and l is the number of training samples. The objective of SVR is to find the best function from a set of possible functions in the form:

$$\{f \mid f(\mathbf{x}) = \mathbf{w}^T\mathbf{x} + b, w \in \Re^d, b \in \Re\} \tag{1}$$

where w is the weight vector estimated by minimizing the regularized risk function showed in Eq. (2) and $\mathbf{b}$ is a bias or threshold

$$\frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{l} L(y_i, f(\mathbf{x}_i)) \tag{2}$$

In order to find the best function f, it is necessary to minimize the regularized risk, where $C > 0$ is a regularization factor, $\|\cdot\|$ is a 2-norm and $L(\cdot,\cdot)$ is a loss function. In order to induce sparsity in SVR the ϵ-insensitive loss function is presented in Eq. (3), which creates an ϵ-tube allowing some predictions to fall within the boundaries of this ϵ-tube:

$$L(y, f(\mathbf{x})) = \begin{cases} 0 & |f(\mathbf{x}) - y| < \epsilon \\ |f(\mathbf{x}) - y| - \epsilon & \text{otherwise} \end{cases} \tag{3}$$

With the ϵ-insensitive function the SVR is formulated as shown in the following equation:

$$\begin{aligned} &\text{minimize} \quad \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{l}(\xi_i - \xi_i^*) \\ &\text{subject to} \quad \begin{cases} \mathbf{w}^T\mathbf{x_i} + b - y_i \le \epsilon + \xi_i \\ y_i - \mathbf{w}^T\mathbf{x_i} - b \le \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \ge 0, \quad i = 0 \ldots l \end{cases} \end{aligned} \tag{4}$$

where ξ and ξ^* are slack variables used to measure the errors occurred by values outside the boundaries determined by the ϵ-tube.

The use of kernels allows the SVR to perform nonlinear mappings into a higher dimensionality space. The regression function takes the form

$$f(\mathbf{x}) = \sum_{i=1}^{l}(\alpha_i - \alpha_i^*)k(\mathbf{x}_i, \mathbf{z}) + b \tag{5}$$

where α and α^* are Lagrange multipliers and $k(\mathbf{x}_i, \mathbf{x})$ is a kernel function. The Gaussian kernel is widely employed in time series forecasting [23,27] ($k(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(\frac{-\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\gamma^2}\right)$), where γ is a parameter of the Gaussian kernel. SVR can produce an unique global minimum solution [24] and depending on the kernel type it can model linear and nonlinear time series.

2.2. Particle swarm optimization

The PSO is an intelligent search algorithm inspired in the social behavior of individuals. The search process is conducted by each

moving particle which represent a possible solution for the problem. Each particle maintains a record of its best solution in a vector called **pbest**, and the best solution among all particles is called **gbest**. The particles move in the direction of the global best solution and its personal best solution.

To perform movements each particle calculates its velocity and updates its position for each iteration. Consider $pbest_i$, $gbest$ and g_i vectors representing the ith particle personal best solutions, global best solution and current solution. The velocity v_i is updated as follows:

$$\mathbf{v}_i(t+1) = w \times \mathbf{v}_i(t) + c_1 \times rnd() \times (\mathbf{pbest}_i - \mathbf{g}_i(t)) + c2 \times rnd() \times (\mathbf{gbest} - \mathbf{g}_i(t)) \tag{6}$$

where $\mathbf{v}_i \in [-v_{max}, v_{max}]$, $rnd()$ is a random function in the range [0, 1], the c_1 and c_2 constants represent the personal and social learning factors respectively and w is the inertia weight. The new solution is calculated as follows:

$$\mathbf{g}_i(t+1) = \mathbf{g}_i(t) + \mathbf{v}_i(t+1) \tag{7}$$

The discrete PSO was proposed by Kennedy and Eberhart [28] to operate on binary search spaces. Each dimension of particle $\mathbf{x_i}$ is a binary value ($\mathbf{g_i} \in \mathbb{B}^k$).

The calculation of the velocity showed in Eq. (6) remains unchanged, except that $\mathbf{g}_i$, $\mathbf{pbest}_i$ and **gbest** belong to the binary domain and $\mathbf{v}_i$ must be constrained to the [0, 1] interval as well. The constriction is obtained by using a sigmoid function $Sig(\mathbf{v}_i)$, and a new position is calculated as follows:

$$\text{if rnd}() < Sig(\mathbf{v}_i), \text{ then} \mathbf{g}_i = 1; \text{ else } \mathbf{g}_i = 0; \tag{8}$$

where $Sig(\mathbf{v_i}) = \frac{1}{1+e^{-\mathbf{v_i}}}$.

2.3. Hybrid models

The combination of models has become a common practice in the time series forecasting field. Real-world data is often composed by linear and nonlinear patterns, thus it is reasonable to perform a combination between linear and nonlinear models. Such combination is achieved by using a linear model to capture linear patterns in the data and the residual error is considered to be nonlinear. In the literature some approaches have been proposed in order to increase the accuracy of predictions.

Zhang [8] considered a given time series y_t to be a composition of linear and nonlinear patterns as shown in Eq. (9), where L_t is the linear component and N_t is the nonlinear component:

$$y_t = L_t + N_t, \tag{9}$$

In his approach, an ARIMA model was employed to map linear patterns and the residual error e_t was obtained by subtracting the ARIMA predictions $\hat{L}_t$ from the actual data y_t as shown in the following equation:

$$e_t = y_t - \hat{L}_t \tag{10}$$

The residual error produced is assumed to have only nonlinear correlations, thus an ANN is employed to perform nonlinear mappings in the residual data. The final prediction ($\hat{y}_t$) is considered to be the summation of both predictions from the ARIMA ($\hat{L}_t$) and the ANN ($\hat{N}_t$) models, with respect to the linear and nonlinear patterns as shown in the following equation:

$$\hat{y}_t = \hat{L}_t + \hat{N}_t \tag{11}$$

Oliveira and Ludermir [23] also assumed a composition of time series as presented in Eq. (9), however a PSO algorithm was employed to determine the order of the ARIMA model, the selection of lags and the SVR parameters simultaneously. A hybrid PSO with discrete and continuous codifications was used to find the best set of parameters for the system. The number of lags was fixed, and the most relevant lags among the sequence were selected to build the SVR model. Linear and nonlinear patterns were mapped by an ARIMA model and SVR respectively.

Babu and Reddy [12] considered the volatility of a time series as a criteria for its decomposition in low-volatility and high-volatility components. According to [12] if a given stationary time series is truly Gaussian, the ARIMA model could provide a better fit. In agreement with the Jarque–Bera normality test, if a kurtosis value of a time series is 3 then the data is Gaussian and is considered to be low-volatile which is appropriate for the application of an ARIMA model. Sequences which not obtained a kurtosis value of 3 were considered to be high-volatile and were modeled by an ANN. Thus a time series is considered to be a sum of a low-volatility (o_t) component and a high-volatility component (h_t) as is presented in the following equation:

$$y_t = o_t + h_t \tag{12}$$

The kurtosis of a series is calculated as shown in the following equation:

$$kurtosis(y) = \frac{E((y-E(y))^4)}{(E((y-E(y))^2))^2}, \tag{13}$$

A moving average (MA) filter (Eq. (14)) was used to separate the time series y_t in two components such that one is less volatile and the other is highly volatile. The length q of the MA filter is adjusted each iteration until the kurtosis has a value of 3 and the decomposition is achieved. The time series resulting from the application of the MA filter is denoted as y_{ma}:

$$y_{ma} = \frac{1}{q} \sum_{i=t-q+1}^{t} y_i \tag{14}$$

The final forecasting $\hat{y}_t$ would be the sum of prediction from each model as presented in Eq. (15), where $\hat{o}_t$ and $\hat{h}_t$ are the predictions of the low-volatility and high-volatility components respectively

$$\hat{y}_t = \hat{o}_t + \hat{h}_t \tag{15}$$

3. Proposed method

The hybrid method proposed by Zhang [8] assumes that a time series can be decomposed in linear and nonlinear patterns. Babu and Reddy [12] considered the volatility of a time series as a criteria for its decomposition in low-volatility and high-volatility components. The moving average filter used in [12] has some drawbacks such as requiring more storage due to all of the q latest observations and applying same weights to the same latest data. Hence if the order (q) of the model is large, past observations will have the same importance as the latest observations [29]. Moreover the system lacks a refined search for model parameters, which could produce more accurate models.

Statistically, if a time series is Gaussian stationary an ARIMA model is more suitable for time series data [12]. The non-Gaussian component may present a richer data structure [30] and may be considered nonlinear [31]. However non-Gaussian patterns may also be captured by linear AR(1) models with the replacement of the innovation distribution by a non-Gaussian distribution [32] as shown in the following equation:

$$h_t = \phi h_{t-1} + \varepsilon_t \tag{16}$$

where ε_t is some non-Gaussian distribution. In this work we consider that a time series may be decomposed into a low-volatility component o_t and a high-volatility component h_t as shown in Eq. (12), in addition the high-volatility component can be further decomposed into linear a_t and nonlinear patterns b_t (Eq.

(17)). This procedure is also described in lines 2 – 4 in Algorithm 1

$$h_t = a_t + b_t \tag{17}$$

Linear component a_t is modeled as a function of its past values using an AR model of order p called AR(p) shown in Eq. (18) (lines 10 – 14). The errors (b_t) produced by the procedure shown in Eq. (19) are used as the nonlinear input for a SVR model which is going to be a function of its k past values (Eq. (20)) (lines 15–16)

$$\hat{a}_t = f(a_{t-1}, a_{t-2}, \ldots a_{t-p}) \tag{18}$$

$$b_t = h_t - \hat{a}_t \tag{19}$$

$$\hat{b}_t = g(b_{t-1}, b_{t-2}, \ldots, b_{t-k}) \tag{20}$$

The final forecast $\hat{y}_t$ will be the sum of the forecast from each component as shown in the following equation (lines 17–20):

$$\hat{y}_t = \hat{o}_t + \hat{a}_t + \hat{b}_t, \tag{21}$$

where $\hat{o}_t$ is the prediction of the low-volatility component using an ARIMA model (lines 6–9), and $\hat{a}_t$ and $\hat{b}_t$ are the linear and nonlinear predictions respectively, from the high-volatility component.

The decomposition of the time series in terms of volatility is achieved through the use of a simple exponential smoothing filter $s_t = \alpha y_t + (1-\alpha)s_{t-1}$ [29], where α is the smoothing factor and needs to be tuned in order to achieve Gaussian and non-Gaussian series. The low-volatility (Gaussian) component is modeled by an ARIMA(p',d,q'). An AR(p) is applied directly to the high-volatility component and the residual error is considered to be the non-linear component.

Most relevant lags from the error produced in the previous step are selected and then modeled by a SVR model. Time series forecasting is based on past values in order to perform predictions, thus the behavior of past data must be taken into consideration. Let $y_t = \{4, 2, 5, 3, 6, 4, 7\}$ be a time series, the procedure converts a scalar time series into a vector $u_t \in \Re^k$ for some integer k, where k is the number of lags. Let the number of lags $k = 4$ and the lag mask $V = \{0, 1, 0, 1\}$, the prediction of the fifth value (6) will take into consideration four past values, however only two values were selected $\{2, 3\}$. The proposed hybrid system architecture is summarized in Fig. 1.

3.1. Parameter optimization

The proposed hybrid system performs two different optimizations, one for the ARIMA model and the other for the AR and SVR models. In order to find a suitable order for the ARIMA model a step-wise procedure is employed [33] to explore the model space more efficiently.

The optimization of high volatility models is performed through the use of a PSO algorithm in a mixed optimization fashion regarding the discrete and continuous space of the problem. The particle codification is presented in Fig. 2.

The PSO algorithm has to perform two different tasks: (i) to find a suitable value of α in order to produce low-volatility and high-volatility series and (ii) to improve the accuracy of the AR-SVR system with both parameter and lag selection. The performance of each particle is measured through a fitness function which is defined in the following equation:

$$fitness_i = (|3 - kurtosis(\mathbf{y})|) + \frac{1}{m}\sum_{j=1}^{m}(h_t - \hat{h}_t^i)^2 \tag{22}$$

The problem was formulated to be a minimization of the fitness function, in which the minimization of the first term will produce a Gaussian series and the second term calculates the mean square error (MSE) of the high-volatility term.

3.2. Computational complexity

A common practice in complexity theory is to concentrate on the worst-case performance of the algorithm [34].

Algorithm 1. The proposed method.

1: Given a time series y_t, $t \in [1 \ldots n]$
2: **for** $t = 1$ to n **do**
3: $o_t = \alpha y_t + (1-\alpha)s_{t-1}$
4: $h_t = y_t - o_t$
5: **end for**
6: Determine the ARIMA(p',d,q') parameters
7: **for** $t = 1$ to n **do**
8: $\hat{o}_t = \phi_0 + \phi_1 o_{t-1} + \phi_2 o_{t-2}, \ldots, \phi_{p'} o_{t-p'} + \theta_1 \varepsilon_{t-1} - \theta_2 \varepsilon_{t-2}, \ldots, \theta_{q'} \varepsilon_{t-q'}$
9: **end for**
10: Calculate $AR(p)$ parameters
11: **for** $t = 1$ to n **do**
12: $\hat{a}_t = \phi_1' h_{t-1} + \phi_2' h_{t-2} + \cdots + \phi_p' h_{t-p}$
13: $b_t = h_t - \hat{a}_t$
14: **end for**
15: Perform lag selection in the predefined k lags.
16: Perform SVR training with the parameters C, ϵ, γ.
17: **for** $t = 1$ to n **do**
18: $\hat{b}_t = SVR(b_{t-\tau_1}, \ldots, b_{t-\tau_r})$, where $r \le k$
19: Produce final forecast $\hat{y}_t = \hat{o}_t + \hat{a}_t + \hat{b}_t$
20: **end for**

Through the analysis of the Algorithm 1, the exponential smoothing filter has a complexity of $O(n)$, where n is the number of data points in a given time series. The estimation of ARIMA order is carried out by a step-wise approach and the parameters estimation is performed by a MLE procedure over l parameters which has a complexity of $O(nl^2)$ [35], where l is the number of parameters for the ARIMA model, considering that the search process needs υ steps, thus the computational cost will be $O(\upsilon nl^2)$. The estimation of the AR model parameters has a computational complexity of $O(np^2)$, for estimating p parameters using MLE, where p is the order of the AR model. The lag selection procedure has complexity of $O(n)$. The LIBSVM [36] was employed for the implementation of SVR model and using k lags has a complexity of $O(nk)$. Reducing the number of lags, the complexity of the SVR model is decreased. The complexity of the proposed model will be $O(nk + \upsilon nl^2 + np^2)$, given all parameters.

However, the optimization of parameters requires an extra computational effort. The PSO search algorithm has a complexity of $O(MX)$ [37] considering X initial particles and M number of iterations. Each particle will essentially execute the proposed algorithm for M iterations (excluding the ARIMA estimation), producing a estimated complexity of $O(MX(nk + np^2) + \upsilon nl^2)$.

4. Experiments

The optimization of the ARIMA and ETS models was implemented using the Forecast package for **R** [33] and the remainder of the system was implemented in MATLAB. The experiments were performed along with the methods proposed by Babu and Reddy [12], Oliveira

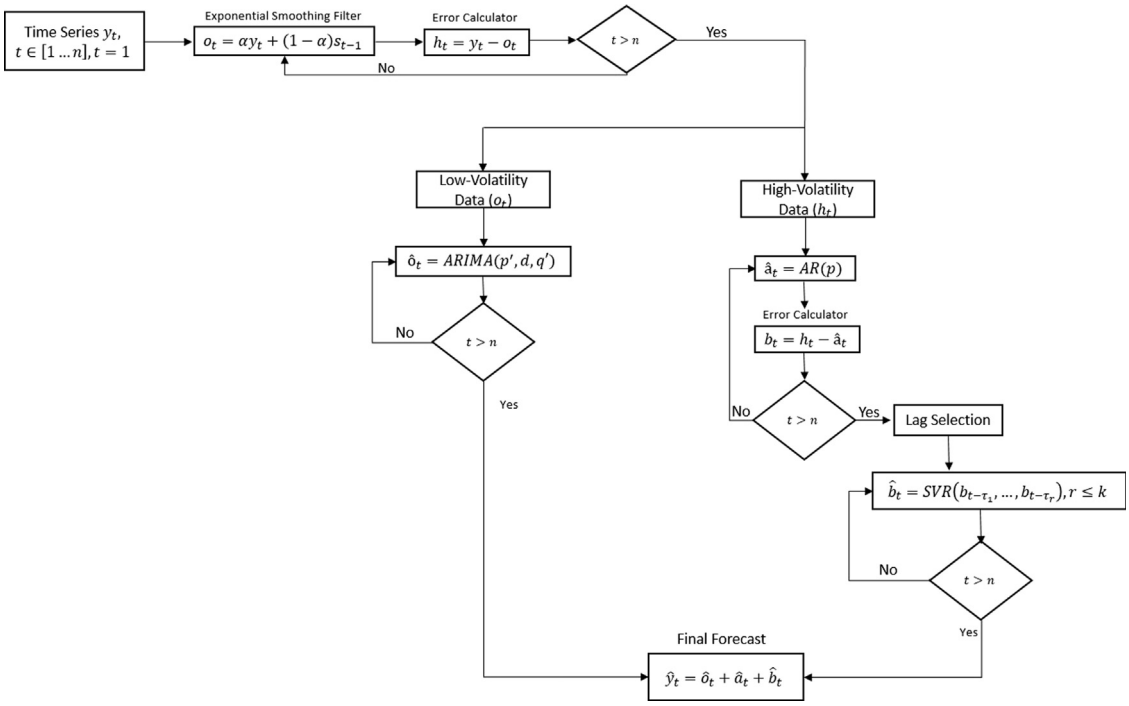

Fig. 1. Proposed hybrid system architecture.

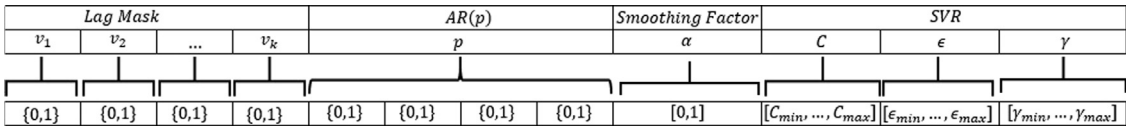

Fig. 2. Particle encodings.

and Ludermir [23], Zhang [8], ARIMA and Exponential Smoothing. In this study, 12 datasets from the Time Series Data Library [25] and one electricity price data from the Australian National Electricity Market [12] were used to assess the performance of the methods. The descriptions of the time series can be found in Table 1. Time series with trend patterns were first detrended and the trend patterns were modeled along with the high volatility component. Two performance measures were employed, which are the mean square error (MSE) and the symmetric mean absolute percentage error (SMAPE) presented in Eqs. (23) and (24):

$$MSE = \frac{1}{m}\sum_{j=1}^{m}(y_t - \hat{y}_t)^2 \tag{23}$$

The MSE is a very popular error measure in the time series forecasting domain, it can be used to guide the training phase of the forecasting methods. However its use is not advised for comparison of forecasting methods across different time series due to some limitations such as being sensitive to outlier data and being scale dependent [38]. The SMAPE has the advantage of being scale independent, providing an easy mechanism to perform comparisons across different time series

$$SMAPE = \frac{1}{m}\sum_{j=1}^{m}\frac{|y_t - \hat{y}_t|}{(|y_t| + |\hat{y}_t|)/2} \tag{24}$$

The number of lags k was determined through the analysis of the autocorrelation function (ACF) of each series. The number of neurons h for the methods of Zhang [8] and Babu and Reddy [12] was determined by performing executions in the range [3, 20] and the results present the configuration that achieved smallest MSE.

Table 1
Datasets description.

Dataset	Description
Passenger	International airline passengers
Stock	Annual common stock price, U.S
Colorado River	Annual Flows, Colorado River Lees Ferry
Dow-Jones	Dow-Jones Industrial Stock Price Index
Electricity	Electricity: electricity end use
IBM	IBM stock: daily
Lake Erie	Monthly Lake Erie Levels
Lynx	Annual number of lynx trapped
NSW	Daily Electricity Price
Pollution	Monthly shipment of pollution equipment
Wine	Monthly Australian wine sales
Sunspot	Wolfs Sunspot Numbers
Accidental death	Monthly Accidental deaths in USA

In the proposed method, the PSO algorithm performs a stochastic search to find the best set of parameters for the SVR model, the order of the AR model, the smoothing factor α and the best set of lags. The PSO algorithm is executed for 100 iterations, the population size was set to 30, the inertia weight (w), personal (c_1) and social (c_2) learning factors assume values stated in the simplified version [5] of the PSO algorithm $w=1$, $c_1=2$ and $c_2=2$ respectively. The search bounds for C, ϵ and γ are within the intervals $[10^4, 10^{-2}]$, $[10^{-1}, 10^{-8}]$ and $[10^3, 10^{-2}]$ respectively.

The methods are evaluated based on one-step-ahead predictions over 30 executions. Also, each series was divided in two subsets: in-samples and out-of-samples. The in-samples were employed in the model building phase, and in the evolutionary approaches this set is further divided into training and validation

sets, whereas the validation set is used to evaluate the model during the evolutionary process. The out-of-samples (test set) has the latest data and is used in the evaluation metrics, and the number of points used in each subset can be found in Table 2, along with the number of lags (k) used. Approximately 80% of the data was used in the in-samples and 20% for out-of-samples. All datasets were scaled in the [0, 1] range using Eq. (25), where $\min_y$ is the minimum value of the series y and $\max_y$ is the maximum value:

$$y' = \frac{y - \min_y}{\max_y - \min_y} \tag{25}$$

Table 2
Datasets division and number of lags.

Dataset	Training	Validation	Test	Number of Lags (k)
Passenger	92	29	23	12
Stock	65	20	14	10
Colorado River	452	149	143	12
Dow-Jones	516	165	143	50
Electricity	298	98	90	12
IBM	227	74	68	10
Lake Erie	366	120	114	12
Lynx	74	23	17	10
NSW	456	149	139	24
Pollution	84	26	20	12
Wine	117	38	32	12
Sunspot	178	58	52	11
Accidental death	48	15	9	12

Table 3
Mean square errors (MSE) for all datasets.

Dataset	Proposed	SVR-LS [23]	Babu [12]	Zhang [8]	ARIMA	ETS
Passenger	0.0016	**0.0013**	0.0020	0.0024	0.0094	0.0100
Stock	0.0073	0.0074	0.0904	0.0106	0.0123	**0.0064**
Colorado River	0.0030	0.0030	0.0029	0.0032	0.0048	**0.0028**
Dow Jones	**0.0005**	0.0014	0.0647	**0.0005**	**0.0005**	**0.0005**
Electricity	**0.0012**	0.0016	0.0024	0.0019	0.0059	0.0117
IBM ($\times 10^{-3}$)	0.6203	0.6072	0.8811	0.7095	0.7282	**0.5495**
Lake Erie	**0.0012**	**0.0012**	0.0018	**0.0012**	**0.0012**	0.0021
Lynx	0.0099	0.0168	0.0097	0.0063	**0.0054**	0.0115
NSW	**0.0018**	0.0020	0.0021	0.0025	0.0024	0.0028
Pollution	**0.0168**	0.0189	0.0215	0.0199	0.0208	0.0344
Wine	0.0201	0.0172	**0.0136**	0.0137	0.0325	0.0402
Sunspot	0.0148	0.0146	0.0147	0.0120	**0.0119**	0.0308
Accidental	**0.0034**	0.0073	0.0207	0.0048	0.0212	0.0215

Table 4
Individual results for the hypothesis test using the MSE. The results indicate which method is better, worse or equivalent to the proposed method using '+', '−' and '≈' respectively.

Dataset	SVR-LS [23]	Babu [12]	Zhang [8]	ARIMA	ETS
Passenger	≈	−	−	−	−
Stock	−	−	−	−	+
Colorado River	≈	+	≈	−	+
Dow Jones	−	−	−	−	+
Electricity	−	−	−	−	−
IBM	≈	−	−	−	+
Lake Erie	−	−	≈	−	−
Lynx	−	≈	+	+	−
NSW	−	−	−	−	−
Pollution	−	−	−	−	−
Wine	+	+	+	−	−
Sunspot	≈	≈	+	+	−
Accidental	−	−	−	−	−

The proposed model has higher computational complexity when compared to the other methods due to the parameter optimization stage. However less human intervention is needed and the predictions are more accurate. In fact, the only *a priori* knowledge about the data is the number of lags k needed for the SVR forecasts.

The analysis is divided in three parts: the first and second are based on the MSE and SMAPE error measures respectively and the last performs comparisons across all datasets.

4.1. Analysis based on MSE

The results with respect to the MSE are presented in Tables 3 and 4. Each value in Table 3 is the mean MSE of 30 executions of the experiment and the lowest values are presented in bold. In order to perform a thorough analysis, a Wilcoxon signed-rank test is applied to compare the proposed method

Table 5
Symmetric mean absolute percentage error (SMAPE) $\times 10^{-2}$ for all datasets.

Dataset	Proposed	SVR-LS [23]	Babu [12]	Zhang [8]	ARIMA	ETS
Passenger	0.0482	**0.0413**	0.0529	0.0586	0.1192	0.1282
Stock	0.1106	0.1122	0.4273	0.1237	0.1315	**0.0951**
Colorado River	0.4799	0.4597	0.4732	0.4988	0.6654	**0.4416**
Dow Jones	0.0244	0.0370	0.3210	0.0251	0.0245	**0.0241**
Electricity	**0.0374**	0.0424	0.0559	0.0489	0.0824	0.1293
IBM	0.1080	0.1079	0.1196	0.1165	0.1188	**0.1020**
Lake Erie	0.0578	0.0571	0.0766	**0.0570**	0.0579	0.0831
Lynx	0.6372	0.7577	0.7469	0.5877	**0.5818**	0.5829
NSW	**0.2943**	0.3135	0.3485	0.3861	0.3769	0.3705
Pollution	0.1868	**0.1854**	0.2036	0.2038	0.2197	0.2561
Wine	0.2287	0.2041	0.1991	**0.1979**	0.2382	0.3150
Sunspot	0.3300	0.3303	0.3450	**0.3289**	0.3602	0.4812
Accidental	**0.0918**	0.1456	0.3055	0.1149	0.2601	0.2645

Table 6
Individual Results for the Hypothesis test using the SMAPE. The results indicate which method is better, worse or equivalent to the proposed method using '+', '−' and '≈' respectively.

Dataset	SVR-LS [23]	Babu [12]	Zhang [8]	ARIMA	ETS
Passenger	+	−	−	−	−
Stock	−	−	−	−	+
Colorado River	+	≈	≈	−	+
Dow Jones	−	−	−	−	+
Electricity	−	−	−	−	−
IBM	≈	−	−	−	+
Lake Erie	+	−	+	−	−
Lynx	−	−	≈	+	+
NSW	−	−	−	−	−
Pollution	≈	−	−	−	−
Wine	+	+	+	−	−
Sunspot	≈	≈	≈	−	−
Accidental	−	−	−	−	−

Table 7
Ranks of all methods using MSE and SMAPE.

Method	Rank MSE	Rank SMAPE
Proposed	55.6154	58.1769
SVR-LS [23]	72.9231	66.6974
Babu [12]	104.0872	108.3205
Zhang [8]	81.5282	85.4974
ARIMA	110.8077	117.7308
ETS	118.0385	106.5769

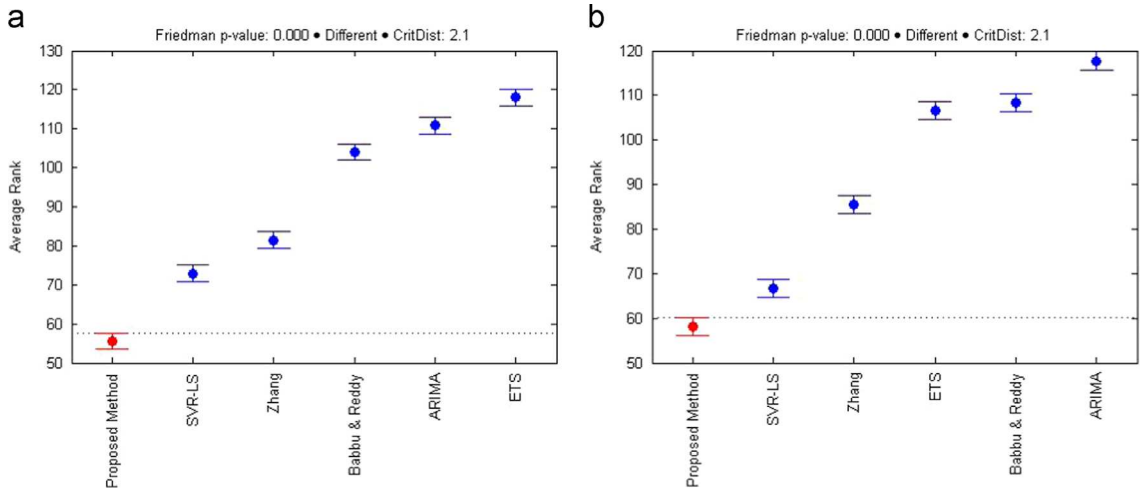

Fig. 3. Hypothesis testing for all datasets in terms of MSE (a) and SMAPE (b).

against the other methods employed in the experiment with a confidence level of 95%.

The results are presented in Table 4 indicating which method is better, worse or equivalent to the proposed method using '+', '−' and '≈' accordingly.

In accordance with the results, the proposed method outperformed all techniques in four cases (Electricity, NSW, Pollution and Accidental). The proposed method achieved better forecasts when compared to: SVR-LS [23] in eight cases; Babu [12] nine cases; Zhang [8] eight cases; ARIMA 11 cases; ETS nine cases.

4.2. Analysis based on SMAPE

The results are also analyzed under the SMAPE criterion which is very important in the time series forecasting domain. The same analysis performed in Section 4.1 is conducted using the SMAPE criterion. Table 5 presents the mean SMAPE over 30 executions of the methods. Employing an empirical analysis the proposed method only achieved lowest errors on three datasets, however when statistically compared with the other techniques some results are similar.

Table 6 shows that the proposed technique achieved lower SMAPE than the other techniques in most datasets.

4.3. Final analysis

Models were evaluated using Friedman and Nemenyi hypothesis tests [39]. Friedman's test was used to rank the algorithms for each data set separately and if the null hypothesis is rejected (ranks are significantly different) a Nemenyi test is employed [40] by comparing all algorithms with each other, performing pairwise tests. Consider a group of ω algorithms, μ datasets and a critical value q_α based on a studentized ranged statistics [40], if the average rank of each algorithm differ by at least a critical distance CD (Eq. (26)), we consider that their performance is significantly different

$$CD = q_\alpha \sqrt{\frac{\omega(\omega+1)}{6\mu}} \tag{26}$$

Using the significance value $\alpha = 0.05$, the value for q_α is 2.8497. A group of algorithms $\omega = 6$ is used and $\mu = 13$ datasets are employed, thus the computed critical distance $CD = 2.0911$, in terms of MSE and SMAPE.

Pairwise testing in Nemenyi test is performed by applying the differences of the ranks within each error measure. When comparing the proposed method with the SVR-LS [23] under the MSE, the absolute value of the difference ($dif = 17.3077$) of the ranks must be achieved and then compared with CD. Since $dif > CD$ then we can conclude that their difference is significant. Table 7 shows the ranks for the hypothesis testing.

Each measure ordered by rank is presented in Fig. 3. In general, the proposed method achieved the best results when compared to the other techniques employed in the experiments in both measures (MSE and SMAPE).

5. Conclusion

In this work a hybrid evolutionary decomposition system for time series forecasting was proposed. The system first decomposes the time series with respect to its low and high-volatility patterns with an exponential smoothing filter, and then decomposes the high-volatility term in linear and nonlinear patterns. It is assumed that there are linear and nonlinear patterns in the high volatility terms, thus an AR and SVR models are used respectively. The parameter optimization was performed through the employment of a PSO algorithm in order to produce more accurate forecasts. The proposed model achieved promising results increasing however the computational complexity in comparison with other techniques.

The experimental analysis was conducted using 13 datasets from the literature and six methods were considered: the proposed method, the PSO-ARIMA-SVR-LS [23], Zhang's method [8], Babu and Reddy's method [12] and an ARIMA and exponential smoothing (ETS) methods. The results showed that no method was the best for all datasets and that in some cases a linear model achieved the best results (lowest errors). Considering the datasets analysed, the proposed method obtained the best overall results in terms of accuracy of predictions and demonstrated promising results for future work. Friedman and Nemenyi hypothesis tests were employed to give a robust evaluation of results and indicated that the proposed method achieved the best rank among the methods.

In order to increase performance of the system, deep learning methods could be employed in order to model deep data structures [41]. Also, the exponential smoothing method could be further explored in a decomposition system due to its performance in the experiments.

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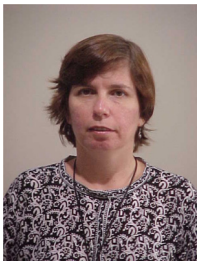

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